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Automated Compliance Mapping for IoT and AI Security Standards: An NLP-Based Approach for ETSI EN 303 645 and EN 304 223

A case study

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Summary

Organizations building AI-enabled IoT products must show compliance with several security standards at once, and aligning their overlapping requirements by hand is slow and demands expertise in both domains. This thesis examines how far that alignment can be automated for two ETSI standards: EN 303 645 for consumer IoT security and EN 304 223 for AI system security. A pipeline was built that extracts 141 normative provisions from the two standards (69 and 72 respectively) and maps them to each other with eleven methods in five families: TF–IDF similarity and security-keyword matching; three sentence-embedding models (Sentence-BERT, MPNet, BGE); three contextual-embedding models pooled from masked language models (BERT, SecureBERT, CySecBERT); an NLI cross-encoder that reads each pair in both directions; Gemini text embeddings; and a Gemini LLM prompted to classify each of the 4,968 possible pairs into one of six relationship types. Predictions were evaluated against a 200-pair reference set built by LLM-assisted annotation: every label drafted by one frontier model, verified in the later two rounds by a second model from an unrelated developer, and reviewed by the authors, drawn from 1,027 pairs judged across three rounds. Related and unrelated pairs are equally represented in it, which fixes the score of the trivial "everything is related" classifier at F1 0.667 and makes that the bar every method must clear.

Four of the eleven methods do not clear it. At the thresholds they ship with, SecureBERT, CySecBERT, Gemini embeddings and, to within one pair, BERT mark every annotated pair as related and reproduce the trivial score exactly, because general-purpose embedding spaces score nearly all provisions as highly similar. Re-selecting each threshold by cross-validation lifts Gemini embeddings to F1 0.78 at precision 0.77, a margin over the baseline that a paired bootstrap separates from zero, with MPNet the only freely available local model that cannot be told apart from it. Calibration is worth having, though at this base rate it buys most methods less than the initial collapse suggests. The domain-adapted cybersecurity models fare worst. SecureBERT lands exactly on the baseline even recalibrated, and CySecBERT, adapted independently by another group, finishes below the general-purpose BERT both were built from. Re-scoring every local method on normative sentences alone, which removes 55% of one standard's words, moves no method's F1 by as much as 0.03, so that collapse is a property of the embedding spaces and not of the explanatory prose they were given. Assigning the correct relationship type is not solved by any of the eleven. The best reaches 0.51 accuracy, matching but not exceeding the majority-class baseline. A control trained on the 827 judged pairs held out of the evaluation, using nothing but the scores those same methods produce, reaches 0.58, so the typed signal is present in the representations and the fixed similarity bands fail to extract it; the overlap/complementarity boundary defeats the trained model as well. Two further findings concern the data rather than the models. Equivalence and subsumption are close to absent between these standards: 13 pairs among the 1,027 judged, and none among 300 top-ranked pairs annotated specifically to find more, so the typed taxonomy is effectively a three-class problem here. The LLM classifier's labels also agree with themselves at only $\kappa = 0.02$ across two runs on identical input, so its reported scores are not stable properties of the method. Because the reference set is balanced by design while the candidate space is sparse, the figures reported are upper bounds on corpus-scale performance. The results support automated mapping as a candidate-ranking aid for human reviewers, not as a replacement for expert judgement.

Affirmation

This master degree report, *Automated Compliance Mapping for IoT and AI Security Standards: An NLP-Based Approach for ETSI EN 303 645 and EN 304 223*, was written as part of the master degree work needed to obtain a Master of Computer Science with specialization in Cybersecurity degree at University West. All material in this report that is not our own is clearly identified and used in an appropriate and correct way. The main part of the work included in this degree project has not previously been published or used for obtaining another degree.

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1 | Introduction

Industrial automation is shifting from traditional rule-based control towards systems that incorporate artificial intelligence and, increasingly, agentic behaviour [1]. At the same time, the number of connected IoT devices keeps growing, and each device widens the attack surface: weak authentication, unprotected update channels, and exposure of personal data are recurring problems [2, 3]. Standardization bodies have responded with security baselines for these technologies. Two of them are central to this thesis. ETSI EN 303 645 defines baseline cybersecurity provisions for consumer IoT devices, covering areas such as password policy, secure communication, software updates, and protection of personal data [4]. ETSI EN 304 223, published by the ETSI committee on Securing Artificial Intelligence, defines baseline security requirements for AI models and systems across their life cycle, from secure design to secure end of life [5].

A company that ships an AI-enabled IoT product falls under both documents. Demonstrating compliance then means working out how the two sets of requirements relate. Some provisions say essentially the same thing; others overlap only partially, or cover different aspects of one shared objective, and many have no counterpart at all. Both standards express their requirements in natural language, at different levels of abstraction and with domain-specific vocabulary, so this mapping work is done manually today. It is slow work, it is error-prone, and it requires expertise in both IoT and AI security [6, 7]. The burden grows with every additional regulation, as the EU Cyber Resilience Act and the AI Act add their own requirement catalogues on top of the sector standards [8].

Recent progress in natural language processing offers a way to reduce this effort. Embedding models built on transformers can measure semantic similarity between sentences even when the wording differs [9, 10], and large language models can be prompted to judge the relationship between two requirements directly [11]. Whether these techniques are accurate enough for security standards, where precision and traceability matter, is still an open question. This thesis investigates it.

1.1 Motivation

Compliance mapping sits in an awkward middle ground. It is too language-heavy for classical rule-based tooling, yet too consequential to hand over to an opaque model without evidence of its reliability. Published work on automated compliance analysis reports promising results in neighbouring domains such as privacy policies, GDPR data processing agreements, and cloud security posture [11, 12, 13], but the specific task of mapping one security standard onto another has received little systematic evaluation. By implementing a spectrum of methods, from keyword matching to LLM classification, and scoring them against a common annotated reference on a real standard pair, this project produces evidence about what automation does well on this task, what it does badly, and what still needs a human. The results are relevant both to researchers working on NLP for requirements engineering and to practitioners who need to decide whether AI-assisted compliance tools can be trusted in their processes.

1.2 Aim

The project builds and evaluates an NLP-based pipeline that identifies semantic relationships between the provisions of ETSI EN 303 645 and ETSI EN 304 223. The aim is not to develop new model architectures but to establish, empirically, how well existing techniques perform on this task. Concretely, the project (i) extracts the normative provisions of both standards into a structured dataset, (ii) applies eleven automated mapping methods spanning rule-based, embedding-based, and LLM-based approaches, (iii) evaluates them against a ground truth of annotated provision pairs that was LLM-drafted, author-reviewed, and cross-verified by a second independent model in two of the three annotation rounds, and (iv) analyses the errors to characterise the limitations of each method family. The outcome is a grounded assessment of the feasibility of automated compliance mapping between IoT and AI security standards, together with a reusable dataset and evaluation framework.

1.3 Research questions

RQ1. To what extent can NLP and AI techniques automatically identify semantic relationships (equivalence, overlap, subsumption, complementarity) between security requirements in ETSI EN 303 645 and ETSI EN 304 223, and what are the fundamental challenges and limitations?

RQ2. How do different automated approaches compare on the compliance mapping task: lexical baselines (TF–IDF, security-keyword Jaccard), sentence-embedding models (Sentence-BERT, MPNet, BGE), contextual-embedding models (BERT, SecureBERT, CySecBERT), a cross-encoder trained for natural language inference, and hosted API methods (Gemini embeddings and Gemini prompt classification)?

RQ3. How does automated mapping compare to human judgement in terms of accuracy, coverage, and efficiency? The comparison is made against the annotated reference set of this project, which was LLM-drafted, cross-verified and author-reviewed rather than built by external domain experts; what that substitution costs the answer is set out in Sections 3.4 and 5.11.

1.4 Contributions

The main contributions of this degree project are:

- A structured dataset of 141 normative provisions extracted from ETSI EN 303 645 and ETSI EN 304 223, and a reference mapping of 200 pairs annotated with six relationship types, justifications, and confidence scores, drawn from 1,027 pairs judged across three annotation rounds and balanced so that the trivial "everything is related" classifier does not score near the top of the scale.
- An open, reproducible pipeline implementing eleven mapping methods (TF–IDF, security-keyword Jaccard matching, Sentence-BERT, MPNet, BGE, BERT, SecureBERT, CySecBERT, an NLI cross-encoder, Gemini embeddings, and Gemini LLM classification) behind a common evaluation harness.
- An empirical comparison of these methods on pair detection and relationship classification, including confidence intervals, threshold sensitivity analysis, and qualitative error analysis, together with two controls that separate the methods from their setup:

an ablation that re-scores every local method on normative text alone, and a classifier trained on the judged pairs held out of the evaluation.

- Evidence-based observations on why general-purpose embedding models overestimate similarity between security requirements, why cybersecurity-specific pretraining did not help in either independent attempt tested, and on the practical role automated mapping can play in compliance work today.
- The finding that equivalence and subsumption are close to absent between these two standards: 13 pairs among the 1,027 judged, and none among 300 top-ranked pairs searched specifically to find more. This bounds what any typed mapping between a device-level and a lifecycle-level standard can report.

1.5 Outline

Chapter 2 introduces the two standards and reviews related work on automated compliance analysis. Chapter 3 describes the research design: data extraction, the relationship taxonomy, ground-truth construction and validation, the mapping methods, and the evaluation metrics. Chapter 4 presents the implemented pipeline. Chapter 5 reports and discusses the experimental results, and Chapter 6 concludes and outlines future work.

2 Background and Related Work

This chapter gives the technical and regulatory context for the project. It opens with the two standards and the security problems they respond to. Compliance mapping as an activity comes next, then the NLP and LLM techniques the project builds on, and the chapter closes on prior studies of automated compliance analysis.

2.1 Background

2.1.1 IoT security and ETSI EN 303 645

Consumer IoT devices are network-connected by design, often built on constrained hardware, and frequently shipped with weak defaults. Recurring weaknesses include poor authentication mechanisms, unprotected software updates, unencrypted communication, unnecessarily exposed interfaces, and insufficient protection of personal data [14, 15]. Because such devices sit in homes and workplaces, even small security failures translate into concrete privacy and safety risks [16].

ETSI EN 303 645 (*Cyber Security for Consumer Internet of Things: Baseline Requirements*) addresses these weaknesses with thirteen groups of provisions, including no universal default passwords, vulnerability disclosure, keeping software updated, secure communication, minimising attack surfaces, and ensuring the protection of personal data [4]. The standard is outcome-focused: it states what a device must achieve rather than prescribing implementations, and it distinguishes mandatory provisions ("shall") from recommendations ("should"). Its provisions are organised in numbered clauses (e.g., provision 5.3-3 on update mechanisms), which makes it a practical anchor for systematic mapping.

2.1.2 AI system security and ETSI EN 304 223

AI systems face threats that classical software security does not fully cover: training data poisoning, model manipulation and theft, adversarial inputs, and prompt injection in systems built on large models [17, 18]. Security work for AI must therefore address the model life cycle (data collection, training, deployment, maintenance, and disposal) and not only the running system.

ETSI EN 304 223 (*Securing Artificial Intelligence: Baseline Cyber Security Requirements*) structures its provisions along exactly that life cycle: secure design (clause 5.1), secure development (5.2), secure deployment (5.3), secure maintenance (5.4), and secure end of life (5.5) [5]. Within these phases it covers AI-specific control areas such as threat and risk management, documentation and auditability of models, protection of datasets and prompts, supply-chain security, security testing, behavioural monitoring, and secure disposal of data and models. The provision-based, life-cycle structure parallels the clause organisation of ETSI EN 303 645, which is what makes a pairwise mapping between the two documents feasible in the first place.

2.1.3 Compliance mapping

Compliance mapping is the activity of relating the requirements of one framework to those of another, so that evidence collected for one can be reused for the other and gaps become visible. Studies of security standards report that overlapping controls, inconsistent terminology, and fragmented requirements are common across frameworks, and that mapping them requires interpretation of purpose and security objective, not just wording [7, 19]. Similar observations come from the GDPR domain, where manual compliance checking is described as time-consuming and dependent on scarce legal-technical expertise [12, 20]. Castellanos Ardila et al. [21] likewise find that manual compliance checking of software processes is labour-intensive and hard to scale.

Automated approaches try to reduce this effort by extracting requirements from the documents and comparing them algorithmically, classifying pairs as identical, related, partially overlapping, or unrelated [22]. Beyond saving effort, automation promises consistency: the same criteria are applied to every pair, which manual mapping cannot guarantee. The open question is accuracy, and that is the subject of this thesis.

2.1.4 NLP and LLM techniques for requirement analysis

Semantic similarity. Standards frequently express similar objectives with different terminology: device-focused language in ETSI EN 303 645, life-cycle and governance language in ETSI EN 304 223. Keyword matching fails on such vocabulary mismatch, a problem long recognised in requirements traceability research [23, 24]. Semantic approaches map text into vector spaces where related meanings land close together, and have been shown to outperform lexical matching for requirement tracing [23].

NLP pipelines. NLP has been applied across requirements engineering tasks: extracting domain concepts, detecting language defects, and establishing traceability links [25, 26]. A typical pipeline extracts normative statements, normalises the text, and converts each requirement into a numerical representation suitable for comparison. Legal and regulatory text adds difficulties of its own: context-dependent terminology, implicit meaning, and complex cross-references [27].

Transformer models. Transformer architectures learn contextual representations of language through self-attention. BERT [9] provides bidirectional contextual embeddings that transfer well to many NLP tasks. For sentence-level comparison, Sentence-BERT [10] fine-tunes a Siamese architecture so that cosine similarity between sentence embeddings reflects semantic relatedness, making large-scale pairwise comparison cheap. SecureBERT [28] adapts a RoBERTa-based model to cybersecurity text, motivating its inclusion here as a domain-tuned alternative. One caveat known from the literature is that the embedding spaces of contextual models are anisotropic: vectors concentrate in a narrow cone, so unrelated sentences can still receive high cosine similarity [29]. This property matters for the results in Chapter 5.

Large language models. LLMs can be prompted to compare two requirements directly and output a relationship judgement, capturing nuances that similarity scores flatten. Reported drawbacks include computational cost, limited reproducibility, and sensitivity to prompt design [30, 11].

Gap analysis. Mapping two standards also serves gap analysis: identifying objectives covered by one document but not the other. ETSI EN 303 645 concentrates on device-level baseline security while ETSI EN 304 223 concentrates on the AI life cycle, so their relationship is expected to consist mostly of partial overlaps and complementary provisions rather than one-to-one equivalences [4, 5]. The task definition therefore covers partial and absent alignment as well as strong matches.

2.2 Related work

2.2.1 Manual comparative mapping of security standards

The closest published work to this thesis is manual. Djebbar and Nordström [7] compare ISO/IEC 27001:2022, ISA/IEC 62443-3-3:2019 and ETSI EN 303 645, and their method is explicitly interpretive: controls are matched by *teleological interpretation*, that is, by the goal a control is intended to realise rather than by the words it uses, and the authors state a deliberate preference for under-mapping over over-mapping when a match is uncertain. Their taxonomy is binary: two controls are *Equivalent* when wording and purpose coincide, and *Related* when the wording differs but the security objective is the same.

Their results are the reference point for what a careful manual mapping of ETSI EN 303 645 produces. All 68 provisions of ETSI EN 303 645 could be aligned with 29 controls of ISO/IEC 27001:2022, while 64 of that standard's 93 controls had no counterpart in ETSI EN 303 645 at all, almost entirely the organizational and people controls, which have no meaning for a device. For ISA/IEC 62443-3-3 they report 84 of 100 requirements mapped, and tabulate the unmapped remainder individually rather than reporting only a coverage figure. The complete mappings are given as appendix tables, one row per aligned control pair.

Two features of that study shape this one. The first is the sparsity: a standards pair drawn from different domains produces far more non-matches than matches, and a mapping method that does not reproduce that asymmetry is not reproducing the task. The second is the form of the output: a typed relation, a coverage statement, and an explicit list of what was left unmapped. Section 5.7 produces those three artefacts automatically.

2.2.2 Automated compliance mapping in cybersecurity

Bar-Haim et al. [13] automate the assessment of organizational cybersecurity posture in cloud environments by mapping free-text security requirements to two control frameworks: a NIST-based framework of 280 controls in 22 families, and a finer-grained three-level risk framework of 169 categories. They formulate mapping as supervised text-pair classification with RoBERTa and SBERT models and use hard negative sampling to force the models to separate semantically close but non-equivalent controls. Their best combined model reaches about 72% precision@1 for control-family mapping and 55% for individual controls, degrading as granularity increases. The authors conclude that the approach suits a human-in-the-loop workflow rather than autonomous operation, a conclusion this thesis revisits for the standard-to-standard setting.

2.2.3 NLP and semantic techniques for requirement mapping

Sapkota et al. [31] present RegCMantic, a framework that maps regulatory guidelines to organizational processes. Regulatory documents are parsed into a uniform XML structure, requirement entities (subject, obligation, action) are extracted into an ontology, and mapping uses three similarity measures over topics, core entities, and auxiliary entities. In a pharmaceutical case study the extended framework improves extraction F-measure from 0.83 to 0.91 and performs mapping evaluated against expert-created references at a 0.85 similarity threshold. The study demonstrates that ontology-supported NLP can automate requirement mapping, while noting that ambiguity and implicit, domain-specific meaning remain the hard part. The same difficulties appear with the standards studied here.

2.2.4 LLM-based compliance analysis

Rodríguez et al. [11] investigate LLMs for large-scale privacy policy analysis, comparing ChatGPT and Llama 2 against earlier symbolic and machine-learning baselines on the MAPP and OPP-115 datasets. With a two-shot prompt design their best configuration exceeds 93% F1, outperforming prior techniques while requiring less annotated data and engineering effort. The study is limited to privacy policies, but it provides strong evidence that prompted LLMs can support compliance-oriented text analysis, motivating the inclusion of an LLM classification method in this project.

2.2.5 Positioning of this thesis

Prior work automates mapping between requirements and control frameworks [13], between regulations and business processes [31], and classification within privacy policies [11]. Mapping two security standards to each other, with a typed relationship taxonomy instead of a binary match, has not to our knowledge been evaluated systematically for the IoT/AI standard pair.

The relationship to [7] is the one worth stating plainly, because this thesis attempts by machine what that study did by hand. It takes the same interpretive task to a standard pair that study does not cover, ETSI EN 303 645 against ETSI EN 304 223; it refines the Equivalent/Related distinction into six typed relations, so that partial and directional alignment can be expressed instead of collapsed; and it asks a question a manual study cannot answer about itself, namely how much of that judgement survives automation and at what error rate. Where the manual study reports its mapping, this one reports a mapping together with a measurement of how often it is wrong, scored against an annotated reference set on which the trivial answer is held to a known score. The comparison of eleven methods that follows measures how closely each approximates the judgement [7] exercised directly.

3 | Research Methodology

3.1 Research design

The project follows an experimental, comparative design. Rather than proposing a new algorithm, it measures how well existing NLP and LLM techniques perform on one concrete task: identifying and typing semantic relationships between the provisions of ETSI EN 303 645 and ETSI EN 304 223. All methods are run on the same extracted dataset and scored against the same ground truth, so differences in the results can be attributed to the methods themselves.

The design addresses the research questions directly. RQ1 (feasibility and limits of automated relationship identification) is answered by the absolute performance levels and the error analysis. RQ2 (comparison of method families) is answered by evaluating all eleven methods under identical conditions. RQ3 (automated vs. human mapping) is answered by treating the reference annotation as the standard of comparison and additionally measuring how consistently the task is performed on repetition. The expert role in that comparison is filled by annotation drafted by one frontier LLM, verified in two of the three rounds by a second from an unrelated developer, and reviewed by the authors against a written codebook, not by external domain experts; a protocol for the external study is given in Appendix B but was not carried out, and Section 5.11 states what this costs the RQ3 answer.

The workflow has five stages: extraction, ground-truth construction, automated mapping, evaluation, and reporting. Each is described in the remainder of this chapter and implemented in the pipeline of Chapter 4.

3.2 Data: standards and provision extraction

3.2.1 Source documents

Both standards are publicly available from ETSI. ETSI EN 303 645 v3.1.3 defines baseline security provisions for consumer IoT devices; ETSI EN 304 223 v2.1.1 defines baseline security requirements for AI models and systems. The two documents differ in structure and terminology but share a numbered-provision format, which the extraction step exploits.

3.2.2 Extraction of normative provisions

Security standards mix normative requirements with explanatory prose, examples, and notes. The PDF documents are converted to plain text with `pdftotext`, recurring page headers and footers are stripped, and provisions are located by their clause identifiers (e.g., "Provision 5.3-3" in ETSI EN 303 645, numbered requirement clauses in ETSI EN 304 223). For each provision the pipeline records: the provision identifier, source standard, section number and title, the provision text, and its modality, that is whether the provision uses *shall* (mandatory), *should* (recommended), or *may* (permitted).

What counts as the provision text needs stating precisely, because the two standards format their requirements differently and the choice affects every similarity score that follows. Blocks the standards label as NOTE or EXAMPLE, along with cross-reference lists and annex material, are removed. ETSI EN 303 645 additionally prints one or more *unlabelled* explanatory paragraphs beneath many of its provisions, giving the reasoning behind the requirement, and these are retained as part of the provision text. The effect is substantial and is reported here rather than left implicit: 25 of the 69 ETSI EN 303 645 provisions carry such a trailer, and the retained explanation accounts for 53% of that standard's extracted words. Provision 5.6-1, for instance, contributes a 19-word requirement followed by 135 words explaining why unused interfaces are disabled. ETSI EN 304 223 has no equivalent construction; its provisions are normative throughout, which is why the two documents are handled by separate parsers.

Retaining the explanation gives the encoders the context in which a short requirement is meant to be read, which is the case for keeping it. The cost is that the explanatory passages are written in general security vocabulary, so they add wording that two otherwise unrelated provisions can share. Section 5.11 treats that as a threat to the internal validity of the similarity results.

The extraction yields 69 provisions from ETSI EN 303 645 and 72 from ETSI EN 304 223, 141 in total. Since any provision of one standard could in principle relate to any provision of the other, the mapping space contains $69 \times 72 = 4{,}968$ candidate pairs. Extraction correctness was checked by sampling provisions and comparing them against the original PDFs; counts per section were also compared with the tables of contents of both standards.

Text preprocessing is deliberately minimal. The transformer models used in this project expect natural sentences and apply their own tokenisation, so provisions are passed to them essentially as extracted. Only the TF–IDF baseline applies lowercasing and English stop-word removal as part of its vectoriser. Domain terms (authentication, encryption, integrity, vulnerability, and so on) are never stripped, since they carry the meaning the mapping depends on.

3.3 Relationship taxonomy

Every candidate pair (a, b), with a from ETSI EN 303 645 and b from ETSI EN 304 223, is assigned exactly one of six relationship types:

EQUIVALENCE
: a and b express the same security objective with comparable scope.

SUBSUMPTION (A BROADER)
: a covers the objective of b and more; b is a special case of a.

SUBSUMPTION (B BROADER)
: the converse: b covers a.

OVERLAP
: the provisions share part of their objective, but each also covers ground the other does not.

COMPLEMENTARITY
: the provisions address different aspects that jointly support a common security goal, without overlapping scope.

No relation
no meaningful connection.

Full definitions with decision guidance appear in Appendix A. For evaluation, the two directional subsumption labels are merged into a single Subsumption class, since direction confusions are less interesting than detecting subsumption at all and the per -direction supports are small.

3.4 Ground-truth construction and validation

3.4.1 LLM-assisted annotation with cross-model verification

Evaluating the automated methods requires a reference mapping. Annotating all 4,968 pairs manually was outside the scope of a 15-credit project, so the ground truth was produced through a staged procedure — LLM drafting, independent cross-model verification for all but the pilot round, and author review — a set-up increasingly common in annotation work and disclosed here for transparency. LLM annotation of text has been reported at or above crowd-worker quality on classification tasks [32], but the same literature is explicit that it should not be trusted unverified: agreement with expert labels varies by task and construct, and human validation of the machine labels remains necessary [33]. The procedure below follows that advice, and adds a second model as an independent judge, a use of LLMs whose agreement with human judgement has itself been characterised [34].

In the first stage, candidate pairs and draft relationship labels were generated with Fable 5 (Anthropic), prompted with the full provision texts and the taxonomy definitions of Section 3.3. In the second, every drafted label in the screened and targeted rounds was re-examined by GPT-5.6 Sol (OpenAI), a frontier model from an unrelated developer, given the same provision texts and codebook but not the first model's justification, and asked to confirm or contest the label. In the third, the authors reviewed the drafts and, where there was one, the verifier's verdict, adjudicated every pair the two models disagreed on, corrected labels and justifications, and removed pairs whose connection did not hold up against the original standard texts. Each retained pair carries a short written justification and a reviewer confidence score from 1 (uncertain) to 3 (certain).

The pilot round preceded the second stage and carries no cross-model verification: its 107 labels were drafted and then read pair by pair by both authors. The two warrants are complementary rather than nested. The pilot rests on human review of every pair, the later rounds on agreement between models from different developers with author adjudication wherever they diverged, and Section 3.4.4 measures the one against the other. Of the 200 evaluated pairs, 37 come from the pilot and 163 from the verified rounds.

Two independent frontier models and a human adjudication step is a stronger warrant than single-pass drafting: in those rounds a label survives only if two models trained by different organisations on different data agree on it, or if an author settled it. Known failure modes of LLM judges — self-preference for their own outputs and position effects [34] — are constrained here because the judge is from a different developer than the drafter and never sees the drafter's justification. What the procedure does not supply is a domain expert [33], and that limit is stated in Section 5.11.

Using an LLM in the annotation loop creates a risk of circularity, since one of the evaluated methods is itself an LLM. Three design choices reduce this risk: neither the drafting model (Anthropic) nor the verifying model (OpenAI) shares a family with the evaluated model (Gemini), the two annotation models are independent of each other, and the labels follow a codebook written before any were drafted, reproduced in Appendix A.

It states the decision procedure, the overlap-versus-complementarity test that carries most of the difficulty, and one amendment made partway through and applied from that point onward.

3.4.2 Three annotation rounds

Annotation proceeded in three rounds, each addressing a limitation of the one before, and the reference set is assembled from their combined output.

The pilot round produced 107 pairs by examining candidates that looked promising and reviewing every draft label. Both authors reviewed these pairs, so they are the closest thing to a human-verified anchor the project has. Their weakness is the same as their construction: pairs chosen because they looked related are 79% related, which tells us about the selection rather than about the standards.

A second, screened round widened the pool. Pairs were ordered by a screening score, the mean percentile rank of each pair across every similarity matrix, and 650 were drawn from three bands of that ordering: 250 from the top tenth, 200 from the next three tenths and 200 from the bottom six tenths. Averaging over the methods keeps the screen neutral between them. Screening on one model's scores would sample densely wherever that model is confident and so flatter it at evaluation. Batches were shuffled under a fixed seed, and the 107 pilot pairs were mixed into the stream unmarked, so re-labelling them was blind and their agreement with the earlier reviewed labels can be read as a check on the new annotation (Section 5.8).

Last, a targeted round went looking for the two classes the first two had barely reached. Across the 727 pairs judged by then, EQUIVALENCE appeared three times and SUBSUMPTION ten. Because those counts cannot support a per-class result, the 300 highest-ranked pairs that no round had yet judged were annotated in eight further batches. They were ranked by the same method-agnostic screen, so no evaluated method is privileged.

That round returned *no* equivalence or subsumption pairs at all: 184 NO RELATION, 65 OVERLAP, 51 COMPLEMENTARITY. This is a property of the two documents and not a shortfall of the search. Among the pairs already judged, the rare classes occur at roughly one in 250 even in the top-ranked region where they concentrate, so reaching twenty of each would require annotating more pairs than the candidate space contains. ETSI EN 303 645 governs a physical consumer device and ETSI EN 304 223 an AI lifecycle and the organizations around it, and a strict-superset or same-scope relation rarely survives that difference in layer. For this pair of standards the typed taxonomy is effectively a three-class problem.

Two consequences are carried forward. Per-class results for EQUIVALENCE and SUBSUMPTION rest on single-digit support and are reported for completeness, not as measurements. And because the rare-class pairs that do exist were partly located by a targeted search, per-class *recall* for those two classes is conditioned on the screen and is not an unbiased quantity; what those rows support is the structure of the confusion, not a recall estimate.

3.4.3 Assembling the reference set

The three rounds leave a pool of 1,027 judged pairs. Because the 107 pilot pairs were judged twice, once in the pilot and once again, unmarked, in the screened round, the pool needs a rule for which of the two labels it carries. It carries the pilot label, on the grounds that those pairs are the only ones two authors examined individually. The consequence is worth stating, because it falls on the rarest classes. Under the pilot labels the pool holds three EQUIVALENCE and ten SUBSUMPTION pairs; under the blind re-annotation of the same pairs it would hold two and four. The rare-class support of the reference set therefore comes almost entirely from the pilot, and the disagreement between the two label sets on those pairs is part of what Section 3.4.4 measures. Neither reading changes the finding that both classes are scarce.

From that pool the evaluation set is drawn under one rule, fixed in advance: *negatives are half the set; positives are balanced across the five positive classes as far as the annotated material allows.*

Each half of the rule does one job. Half negatives keeps the trivial classifier that calls every pair related from scoring near the top of the scale: at a positive rate of one half it scores $F_1 = 0.667$, which every detection table in Chapter 5 carries as its final row. Balancing the positives keeps the typed task from becoming a report on whichever relationship happens to be most common. Where a class cannot meet its share it contributes everything it has and the remainder is redistributed, which the scarcity of the rare classes forces here.

Table 3.1: Composition of the 200-pair reference set. EQUIVALENCE and SUBSUMPTION enter at their full available count rather than at the balanced target, because the corpus contains no more of them.

Label	Judged	In set	
NO RELATION	701	100	half the set by rule
OVERLAP	174	44	
COMPLEMENTARITY	139	43	
SUBSUMPTION (B broader)	6	6	all available
SUBSUMPTION (A broader)	4	4	all available
EQUIVALENCE	3	3	all available
	1 027	200	positive rate 0.500

The realised set is given in Table 3.1: 200 pairs, exactly half related, of which 37 come from the author-reviewed pilot. The over-supplied classes are drawn under a fixed seed, so the set reproduces exactly.

Two questions about this design deserve an answer before the results are read.

The first is why the evaluation uses 200 pairs when 1,027 were judged. Scoring the whole pool would give narrower intervals, but the pool is 68% negative and its positive half is dominated by OVERLAP and COMPLEMENTARITY, so both the detection baseline and the typed baseline would be set by a class balance that no design decision controls. Drawing to a fixed composition makes both baselines known quantities and keeps the typed task from reducing to a report on the most common positive label. The price is statistical power, and the full judged pool ships with the repository so the alternative can be computed.

The second is what the balance costs in generality. A positive rate of one half is a design choice rather than a property of the corpus, and precision and F1 both move with it. Every figure reported in Chapter 5 therefore describes performance on a balanced set and is an upper bound on what the same operating point achieves across the full 4,968 candidates, which are far sparser. Quantifying that gap would require a sample drawn with known inclusion probabilities. This design does not have one, and the future work of Chapter 6 names it as the first extension the work should receive.

3.4.4 Reliability of the reference set

Three statistics are reported, all computable from what was actually done. The first is agreement between the sample and the author-reviewed pilot labels on the pairs they share, using raw agreement and Cohen's kappa [35] with the bands of Landis and Koch [36]. The second is calibration: whether recorded annotation confidence predicts where the two label sets diverge. The third is a contrast rather than a validation: the run-to-run self-agreement of the evaluated LLM on identical input, which shows what an unreliable annotation source looks like on the same scale.

Cross-model agreement using Gemini is deliberately excluded. Gemini is one of the evaluated methods, so its labels cannot serve as an independent check on a reference set used to score it.

A larger validation with external cybersecurity professionals was designed but not executed within the project timeframe; the complete survey instrument and analysis plan are provided in Appendix B as a ready-to-run protocol for future work.

3.5 Automated mapping methods

Eleven methods are evaluated, grouped into five families. The grouping is not housekeeping: it is what the comparison is about. Lexical baselines match words. Sentence-embedding models are trained so that cosine similarity means something. Contextual-embedding models are masked language models pooled into a vector without any such training. A cross-encoder reads both provisions at once and answers a directional question. Hosted API models are the commercial current generation. Differences within a family isolate the effect of the model; differences between families isolate the effect of the approach.

3.5.1 Rule-based baselines

TF–IDF cosine similarity. Each provision is represented as a TF–IDF-weighted bag of words; every cross-standard pair receives the cosine similarity of the two vectors. This captures shared vocabulary and nothing else, making it the canonical lexical baseline.

Security-keyword Jaccard. A curated list of cybersecurity terms (password, authentication, encryption, vulnerability, logging, risk, update, and others) is matched against each provision; a pair is scored by the Jaccard overlap of the keyword sets. This baseline mimics how a checklist-driven analyst might shortlist related provisions.

3.5.2 Sentence-embedding models

Three encoders trained so that cosine similarity between sentence embeddings tracks semantic relatedness. Each provision is embedded once and every pair is scored by the cosine of the two vectors.

- **Sentence-BERT** (`all-MiniLM-L6-v2`) [10], the compact model that made this style of comparison cheap enough to run over a full pair space;
- **MPNet** (`all-mpnet-base-v2`) [37], the strongest general-purpose model of the same family, included so that the effect of model capacity can be separated from the effect of the task;
- **BGE** (`bge-base-en-v1.5`) [38], a retrieval-trained embedder a generation newer than the other two, included as the current state of the open embedding literature.

3.5.3 Contextual-embedding models

Three masked language models with no sentence-similarity training, mean-pooled over token embeddings and L2-normalised. They are included to separate two explanations for embedding performance that are easily confused: the size of the model, and whether it was ever trained to make its geometry mean something.

- **BERT** (`bert-base-uncased`) [9], the general-purpose baseline;
- **SecureBERT** [28], a cybersecurity-domain RoBERTa model;
- **CySecBERT** [39], a second, independently built cybersecurity-domain model. One domain-adapted model that behaves oddly is an anecdote about that model; two, adapted by different groups from different base models, make it a statement about domain adaptation for this task.

3.5.4 Cross-encoder inference

Every method above reduces a pair to one symmetric number, and a symmetric number cannot say which of two provisions is the broader one. Subsumption is therefore unreachable for all of them by construction. The NLI cross-encoder (`nli-deberta-v3-base`, a DeBERTa-v3 model [40] fine-tuned on natural language inference data [41]) reads both provisions together and returns whether the first entails the second, which is a directional question.

Each pair is scored twice, once in each direction, and the two entailment probabilities decide the label: entailment both ways is EQUIVALENCE; entailment one way only is SUBSUMPTION with the entailed provision as the narrower one; some entailment but not enough in either direction is OVERLAP; and no entailment either way is NO RELATION. A direction counts as entailed at probability 0.50, and 0.20 is the floor below which the method emits nothing at all. Both cut-offs were fixed before any evaluation. The stored score is the larger of the two entailment probabilities, so this method can be swept and recalibrated on the same footing as the others. The cost is two forward passes per pair, 9,936 in total, against one embedding pass per provision for the bi-encoders.

3.5.5 Hosted API methods

Gemini embeddings. Each provision is embedded with Google's `gemini-embedding-2` model via API; pairs are scored by cosine similarity, exactly as for the local encoders. This represents the current generation of commercial embedding models.

Gemini LLM classification. For every one of the 4,968 pairs, a prompt containing both provision texts and the six label definitions asks `gemini-2.5-flash-lite` to answer with exactly one label. Jobs are submitted through the batch API. Unlike the similarity methods, this approach predicts a relationship type directly and can, in principle, distinguish overlap from complementarity, which a scalar similarity cannot.

3.5.6 From similarity scores to labels

The similarity-based methods produce scores in $[0, 1]$, not labels. Scores are mapped to relationship types with fixed thresholds: ≥ 0.85 EQUIVALENCE, ≥ 0.70 OVERLAP, ≥ 0.50 COMPLEMENTARITY, otherwise NO RELATION. A second, lower threshold decides whether a pair is emitted as a candidate at all, and it is set per family rather than per model: 0.10

and 0.05 for the two lexical baselines, 0.30 for the sentence-embedding models, 0.70 for the contextual-embedding models, 0.50 for the Gemini embeddings, and the entailment floor of Section 3.5.4 for the cross-encoder. These cut-offs follow the operating points reported in related mapping work [31], and they are fixed per family rather than tuned per model so that a difference between two models in the same family reflects the representation and not the calibration.

That decision has a consequence worth stating in advance rather than discovering in the results. A threshold chosen a priori is a threshold chosen without knowing where a method's scores actually fall, and Section 5.3 shows that several methods occupy ranges their family default does not reach into at all. Those methods therefore ship at operating points that place every pair on one side of the line, which is why every method with a stored score matrix is also reported at a threshold recalibrated on the reference set. The calibration analysis in Section 5.4 shows how much of each method's performance is representation and how much is the choice of operating point.

Subsumption cannot be expressed as a symmetric similarity band at all, so ten of the eleven methods cannot predict it by construction. Only the cross-encoder can, which is the reason it is in the study.

3.6 Evaluation framework

Evaluation is restricted to the 200 annotated pairs; predictions outside the annotated set are ignored, since their true labels are unknown. Two views are computed for every method.

3.6.1 Pair detection

Detection asks whether a method finds related pairs at all, regardless of label. A pair is a positive if its ground-truth label is anything but No relation. With TP, FP, FN counted within the annotated scope:

$$\text{Precision} = \frac{TP}{TP+FP}, \qquad \text{Recall} = \frac{TP}{TP+FN}, \qquad F_1 = \frac{2PR}{P+R}. \tag{3.1}$$

Two consequences of scoping deserve emphasis. False positives can only be counted on the 100 annotated negative pairs, so a method that predicts relations sparsely can show high precision while missing nearly everything; precision values must be read together with recall and with the predicted-positive counts reported alongside. And because the set is half positive by construction, the classifier that calls every pair related scores $F_1 = 0.667$ on it. That figure is reported as a row in every detection table, since a detection score means little except in relation to what the trivial answer achieves on the same pairs.

3.6.2 Relationship classification

Classification asks whether the predicted label matches the ground truth, over all 200 pairs, with the subsumption directions merged (Section 3.3). Ground-truth pairs absent from a method's output count as predicted No relation. Reported metrics are overall accuracy and macro-averaged precision, recall, and F1 over the five classes, plus per-class scores. Macro averaging weights all classes equally, which is appropriate in principle because the rare classes (equivalence, subsumption) are exactly the ones a compliance analyst cares most about. With three and ten supporting pairs respectively, however, those two components move substantially on a single prediction, and the macro average inherits that instability. The majority-class classifier, which predicts No relation for everything, scores 0.500 accuracy and 0.133 macro-F1 on this set and is the reference the typed results are read against.

3.6.3 Uncertainty and sensitivity

With 200 evaluation pairs, point estimates alone would overstate certainty. Nonparametric bootstrap confidence intervals (2,000 resamples of the annotated pairs, percentile method, 95% level) are reported for detection F1 and macro-F1. For every method with a stored similarity matrix, detection metrics are additionally computed across the method's full score range, to separate the quality of the underlying representation from the choice of operating point.

Results are reported at two operating points, because the shipped thresholds were fixed per family before any evaluation and applying them confounds the method with its calibration. The second point is chosen on the reference set itself: F1 is maximised on four folds and scored on the fifth, over 20 repeats, with folds stratified by label. Candidate cut-offs are quantiles of each method's own score distribution rather than a shared absolute grid, since the methods occupy ranges that differ by an order of magnitude in width. TF–IDF tops out at 0.190 while SecureBERT never drops below 0.875, so an absolute grid would give one of them a hundred candidate cut points and the other three. The reported threshold is the median of the 100 selections, so the point estimate that follows is not the in-sample optimum; the gap between it and the cross-validated mean is reported as the residual optimism.

Comparisons between methods use a paired bootstrap on the difference: both methods are scored on the same resample, and only differences whose interval excludes zero are reported as orderings. Two marginal intervals overlapping is not evidence that the methods are equivalent, because the methods are scored on identical pairs and their errors are correlated.

Which comparisons are made is decided by a rule, not by inspection. Eleven methods admit fifty-five pairwise contrasts, and fifty-five uncorrected 95% intervals will contain false findings whatever the data say. Two questions carry the argument: whether a method beats the all-positive baseline, and whether anything beats the best-scoring method. Each method is therefore compared against those two references and against nothing else. One caveat attaches to the second reference. It is selected as the maximum on this sample, and a difference measured against a selected maximum is biased in that maximum's favour, because the selection absorbs part of the sampling noise. Those intervals describe the gap to the leader; they are not a test that the leader is genuinely best.

3.6.4 Efficiency and interpretability

Wall-clock runtime of each stage is recorded on commodity hardware (Apple silicon laptop, CPU only). Interpretability is assessed qualitatively: whether a method's output carries any human-readable rationale (matched keywords, a similarity score, or a generated label) that an analyst could act on.

4 | The Mapping Pipeline

The methods of Chapter 3 are implemented as a five-stage pipeline, released as an open repository so that every number in Chapter 5 can be regenerated from the raw standard documents. This chapter describes the system's structure, what was built for the project and what was reused, and the artefacts each stage produces.

4.1 Architecture

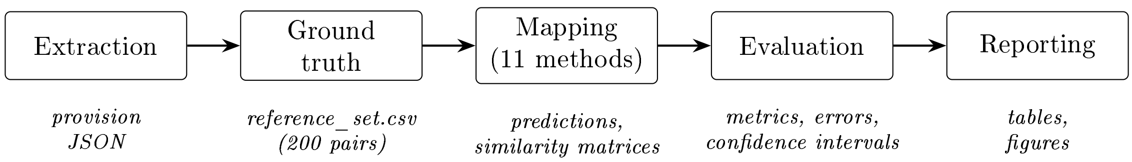

Figure 4.1: Pipeline stages and the artefact each stage produces. Every stage is a separate command, so intermediate outputs can be inspected before the next stage runs.

Figure 4.1 shows the five stages. The pipeline is a set of Python modules run stage by stage from the command line; each stage reads the files of the previous one and writes its results to a versioned data directory. This staged design was a deliberate choice over a single end-to-end script: extraction errors are cheaper to catch before mapping runs, and the API-based methods (which cost money per call) are only invoked once their inputs are verified.

4.2 Stage descriptions

Extraction. Converts both standard PDFs to text with `pdftotext` (Poppler), removes recurring ETSI page furniture, locates provision identifiers with pattern matching, and writes one JSON record per provision with identifier, section, title, text, and modality. Output: 69 records for ETSI EN 303 645, 72 for ETSI EN 304 223.

Ground truth. Materialises the annotation of Section 3.4 as `reference_set.csv`, holding for each pair the source and target identifiers, one of six relationship labels, a one-sentence justification, a confidence score, the round the pair came from, and whether the authors reviewed it. Companion modules run the three annotation rounds (the screened draw, the resumable emit/ingest batch harness, and the targeted search for the rare classes), assemble the evaluated set from their combined output under the composition rule of Section 3.4.3, and compute the agreement statistics of Section 3.4.4.

Mapping. One module per method. The rule-based module computes TF–IDF cosine similarity (scikit-learn) and security-keyword Jaccard scores. The six local transformer modules share a common embedding helper that loads the model (Hugging Face), embeds all provisions, normalises the vectors, and computes the full 69×72 cosine similarity matrix. The cross-encoder module runs its own loop, scoring each pair twice so that the direction of entailment is recorded. The Gemini embedding module obtains vectors through the Google GenAI API; the Gemini classification module builds one prompt per pair and submits them through the batch API, then parses the returned labels. Every similarity-based module stores both its thresholded predictions (CSV) and its raw similarity matrix (NumPy), the latter enabling the threshold sweep of Section 5.4 without re-running any model.

Evaluation. Loads the ground truth and every prediction file, computes detection and classification metrics (Section 3.6), bootstrap confidence intervals, per-method confusion matrices, and threshold sweeps, and writes per-method error files listing every missed pair and every wrong label with its provision identifiers. Two further modules run the controls of Sections 5.3 and 5.6: one re-scores the local methods on provision text stripped to its normative sentences, the other trains a classifier on the judged pairs held out of the reference set using the stored similarity scores as its only features. Both write to separate directories and neither can alter a reported figure.

Reporting. Renders the summary tables and the figures used in Chapter 5 (matplotlib), in print-ready PDF form.

4.3 Built versus reused

The extraction logic, the keyword baseline, the ground-truth tooling, the evaluation harness with its bootstrap and sweep analyses, and the reporting layer were developed for this project. The mapping methods themselves reuse standard components: scikit-learn's TF–IDF vectoriser, seven pretrained models from Hugging Face (Sentence-BERT, MPNet, BGE, BERT, SecureBERT, CySecBERT and the DeBERTa-v3 NLI cross-encoder), and Google's hosted Gemini models. No model was trained or fine-tuned. The contribution is the assembly and the evaluation of existing components under identical conditions.

4.4 Reproducibility

All randomised steps (sampling, bootstrap) use fixed seeds. A verification run during the project reproduced every stored evaluation metric exactly; individual embedding scores vary in the fourth decimal between runs because of CPU thread scheduling in the tensor library, which never crosses a label threshold. The API-based methods depend on hosted models and are archived through their stored outputs instead. Appendix C lists the exact commands, software versions, and expected runtimes for a full reproduction.

5 Results and Discussion

This chapter reports the experimental results. Section 5.1 recalls the setup, Sections 5.2–5.5 present pair detection, similarity structure, threshold calibration, and relationship classification, Section 5.7 turns the best method's output into a coverage statement, Section 5.8 covers the ground-truth validation, and Sections 5.9–5.11 discuss the findings against the research questions and the threats to their validity.

5.1 Experimental setup

All eleven methods were run on the full cross product of 69 ETSI EN 303 645 provisions and 72 ETSI EN 304 223 provisions (4,968 pairs) and evaluated on the 200-pair annotated reference set described in Section 3.4: 100 related pairs and 100 annotated negatives (Figure 5.1). Metrics outside the annotated scope are not computable and are not claimed.

Two properties of that set govern how every number below should be read. The first is the half-and-half split between related and unrelated pairs, fixed by the composition rule of Section 3.4.3. It holds the trivial classifier that calls every pair related to $F_1 = 0.667$, and every detection table below carries that value as its final row. It also means the figures reported here describe a balanced set and bound from above what the same operating points achieve at corpus scale.

The second is the scarcity of the rare classes. OVERLAP (44 pairs) and COMPLEMENTARITY (43) make up almost all the positives, while SUBSUMPTION contributes 10 pairs and EQUIVALENCE 3. As Section 3.4.2 reports, this reflects the two standards rather than the draw: only 13 of the 1,027 judged pairs fall into those classes, and the targeted search for more returned none. Per-class figures for EQUIVALENCE and SUBSUMPTION therefore rest on single-digit support and are reported for completeness rather than as measurements.

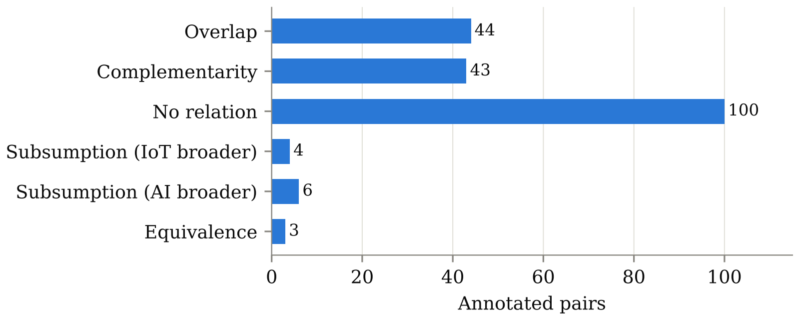

Figure 5.1: Distribution of relationship labels in the 200-pair reference set.

5.2 Pair detection

Table 5.1 and Figure 5.2 show detection at each method's shipped operating point; Figure 5.3 adds bootstrap confidence intervals.

Table 5.1: Pair detection on the annotated scope at each method's shipped threshold. P^+ is the number of pairs the method marks as related among the 200 annotated pairs. The final row is a classifier that labels every pair related.

Method	P^+	TP	FP	FN	Precision	Recall	F1
TF–IDF	2	1	1	99	0.500	0.010	0.020
Keyword Jaccard	11	6	5	94	0.545	0.060	0.108
Sentence-BERT	17	15	2	85	**0.882**	0.150	0.256
MPNet	14	12	2	88	0.857	0.120	0.210
BGE	177	93	84	7	0.525	0.930	**0.671**
BERT	199	100	99	0	0.502	1.000	0.669
SecureBERT	200	100	100	0	0.500	1.000	0.667
CySecBERT	200	100	100	0	0.500	1.000	0.667
NLI cross-encoder	8	7	1	93	0.875	0.070	0.130
Gemini embeddings	200	100	100	0	0.500	1.000	0.667
Gemini LLM	179	88	91	12	0.492	0.880	0.631
Label every pair related	*200*	*100*	*100*	*0*	*0.500*	*1.000*	*0.667*

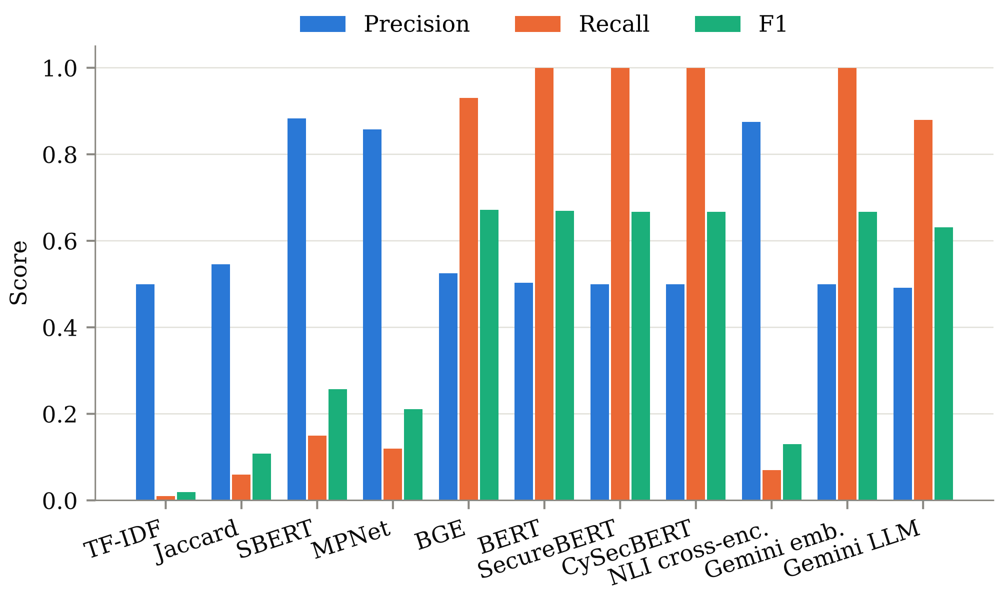

Figure 5.2: Pair-detection precision, recall, and F1 per method.

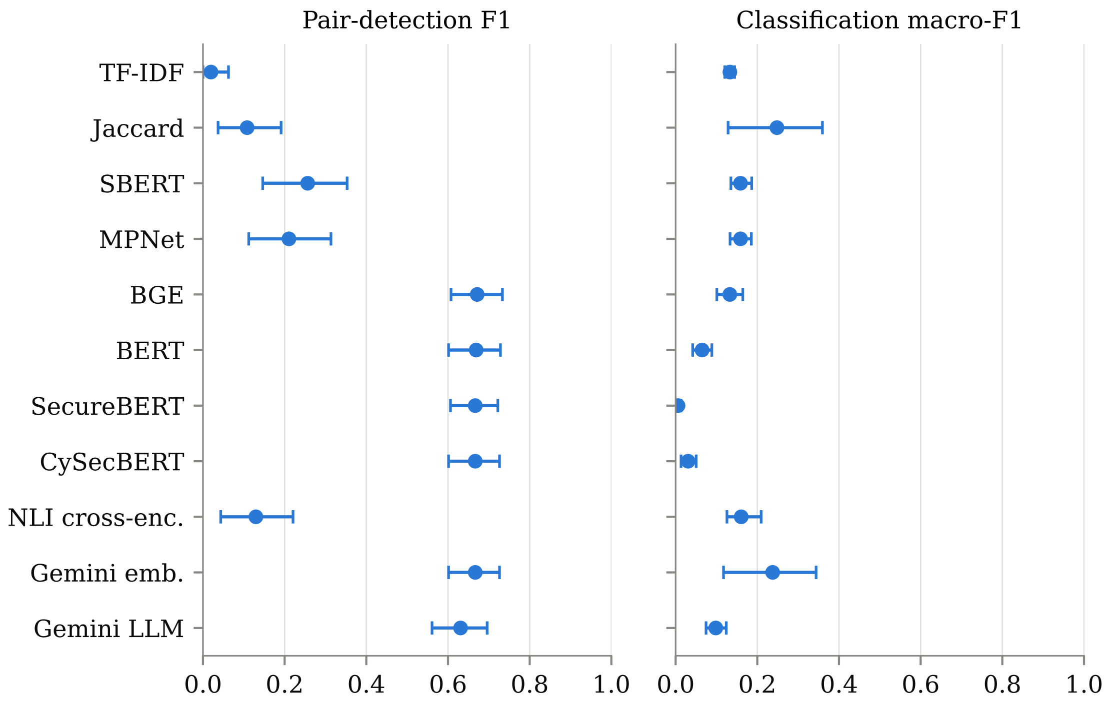

Figure 5.3: Bootstrap 95% confidence intervals (2,000 resamples) for pair-detection F1 and classification macro-F1.

The baseline row splits the methods into two regimes. The lexical baselines, both smaller sentence-embedding models at their default operating points, and the cross-encoder predict very few relations. What they do predict is largely correct, with Sentence-BERT reaching precision 0.882 and the cross-encoder 0.875, but recall between 0.01 and 0.15 makes none of them usable as a stand-alone detector.

The remaining methods sit at the opposite extreme. SecureBERT, CySecBERT and Gemini embeddings mark *every one* of the 200 annotated pairs as related, and BERT marks all but one. Their F1 of 0.667–0.669 is what the all-positive baseline scores on the same pairs, to three decimals. At the thresholds they ship with, these four methods implement the constant function. BGE clears the baseline by 0.004 and the Gemini LLM falls below it. Detection at shipped thresholds separates almost nothing, and the balanced set makes that visible in the table itself.

5.3 Similarity structure

Figure 5.4 explains the degenerate behaviour, and with eleven methods the pattern is legible as a family property and not a quirk of one model. All three contextual-embedding models compress: BERT places every pair, related or not, between 0.570 and 0.914; CySecBERT between 0.707 and 0.925; SecureBERT is the extreme case, packing the entire corpus into $[0.875, 0.971]$, a band 0.096 wide. With ranges that narrow, the fixed label thresholds of Section 3.5.6 land every pair in the same band: SecureBERT assigns EQUIVALENCE to all 4,968 possible pairs, and CySecBERT splits them between EQUIVALENCE and OVERLAP without ever reaching NO RELATION. This matches the known anisotropy of contextual embedding spaces [29]: mean-pooled vectors from models never trained for sentence similarity crowd into a narrow cone, and cosine similarity loses its discriminative

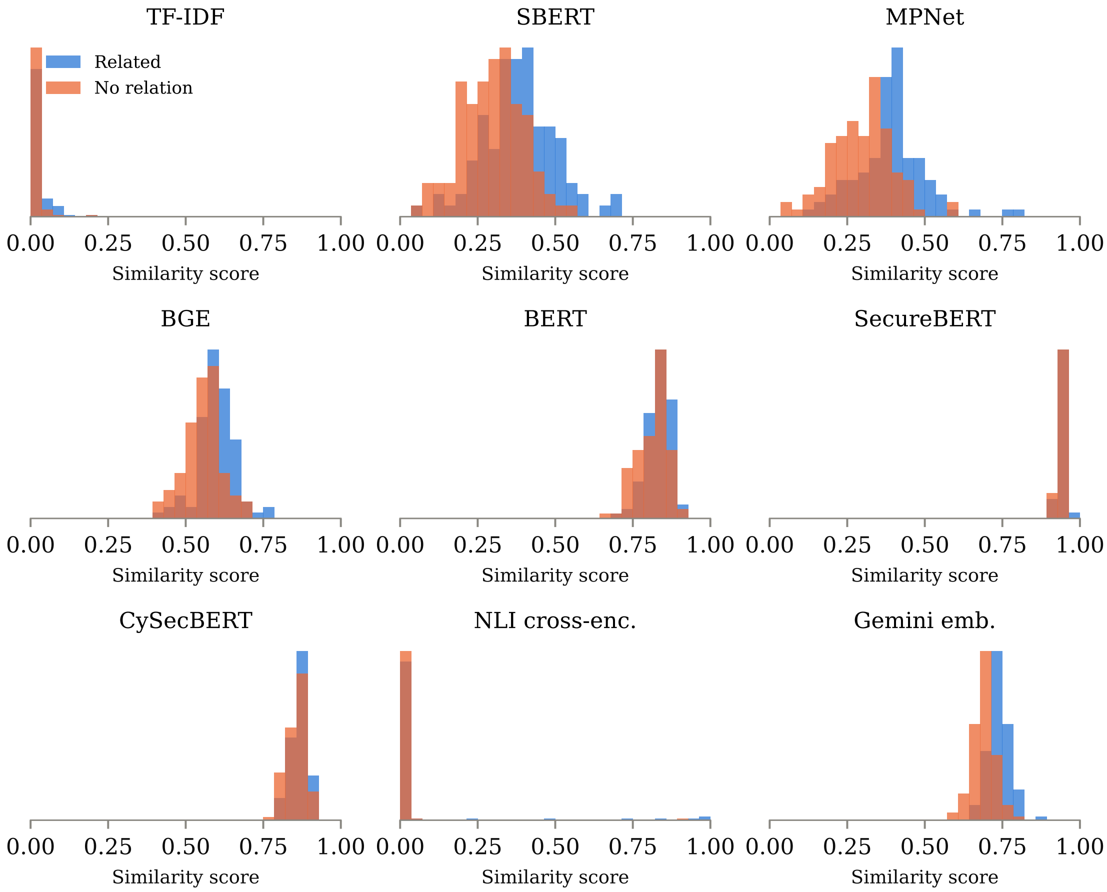

Figure 5.4: Score distributions for annotated related pairs (blue) and annotated negatives (orange), one panel per method with a continuous score. Panels are not on a common scale: what matters is the separation between the two distributions within a panel, not where either sits.

meaning. Adding cybersecurity vocabulary does not repair the geometry: both domain-adapted models are more compressed than the general-purpose model they descend from.

The sentence-embedding models spread out, as their training intends. Sentence-BERT and MPNet span roughly $[-0.05, 0.8]$ and visibly separate the two distributions, though even there the overlap is substantial, which confirms that the annotated negatives are genuinely hard. BGE sits in between. It is trained for similarity and separates the classes well, but its scores never fall below 0.345, so a threshold chosen for its family lands beneath its entire distribution. Its saturation comes from where its scores sit and not from what they encode, and the recalibration in Section 5.4 isolates that distinction. Gemini embeddings have a compressed range of 0.297 but shift related pairs consistently above negatives, which is what lets them lead the calibrated table. The NLI cross-encoder produces a probability rather than a cosine and places the bulk of pairs near zero, which is why its panel looks empty on the right.

Is the compression the model or the input?

The explanation above attributes the compression to model geometry, but the input gives a competing account. As Section 3.2 records, ETSI EN 303 645 prints unlabelled explanatory paragraphs beneath many of its provisions and the extraction retains them, so a large part of the text being encoded is general security prose that unrelated provisions can share. If that is what crowds the scores, the anisotropy argument is not needed to explain them.

The two accounts are separable by re-running the local methods on provision text stripped to its normative sentences. The trailers carry no marker of their own, the extractor having already removed the blocks the standard labels NOTE or EXAMPLE, so they are identified by the absence of a normative modal: sentences containing *shall* or *should* are kept and the rest discarded. The operation removes 55% of ETSI EN 303 645's words across 27 of its 69 provisions and only 7% of ETSI EN 304 223's across 9 of 72, which is the asymmetry Section 3.2 describes and a check that the heuristic is cutting what it is meant to cut. The 55% is close to the 53% that section attributes to the trailers directly; the two are separate operationalisations of the same boundary and need not coincide exactly. Every method is then recalibrated by the same cross-validation used in Section 5.4, so the two conditions are compared at operating points chosen the same way. The two hosted methods are archived as stored outputs and cannot be re-scored, so the comparison covers the nine local ones.

Table 5.2: Each local method re-scored on provision text stripped to its normative sentences, against the full text used everywhere else. Width is the span between the method's lowest and highest score over all 4,968 pairs; F1 is detection at a threshold recalibrated separately in each condition.

	Score range width			Calibrated F1		
Method	Full	Req. only	Δ	Full	Req. only	Δ
TF–IDF	0.190	0.265	+0.075	0.703	0.682	−0.021
Sentence-BERT	0.750	0.748	−0.002	0.700	0.713	+0.013
MPNet	0.863	0.869	+0.006	0.721	0.694	−0.027
BGE	0.420	0.475	+0.056	0.711	0.711	+0.000
BERT	0.344	0.440	+0.096	0.693	0.682	−0.011
SecureBERT	0.096	0.119	+0.024	0.667	0.667	+0.000
CySecBERT	0.218	0.212	−0.007	0.685	0.683	−0.001
NLI cross-encoder	0.993	0.939	−0.054	0.673	0.660	−0.013

Table 5.2 answers the question, and it answers it against the input. Removing more than half of one standard's text moves no method's detection F1 by as much as 0.03, in either direction, and the two changes large enough to notice point opposite ways: MPNet loses 0.027 and Sentence-BERT gains 0.013. The score ranges widen slightly for the compressed models, by 0.096 for BERT and 0.024 for SecureBERT, so the explanatory prose does contribute something to the crowding, but not the phenomenon. SecureBERT still confines the entire corpus to a band 0.119 wide and still scores exactly the all-positive baseline after recalibration. A model whose geometry the input was responsible for would not survive the removal of half its input unchanged.

The compression is therefore a property of the models, and the anisotropy account [29] stands without needing the input to support it. The result also settles a threat the study would otherwise have had to leave open, which is why it is reported here rather than deferred to Section 5.11.

5.4 Threshold calibration

The previous two sections raise one question: when a method scores at the baseline, is that because its representation carries no signal, or because its threshold sits in the wrong place? Figure 5.5 separates the two by sweeping the operating point, and Table 5.3 reports what each method reaches when the threshold is chosen by cross-validation instead of inherited.

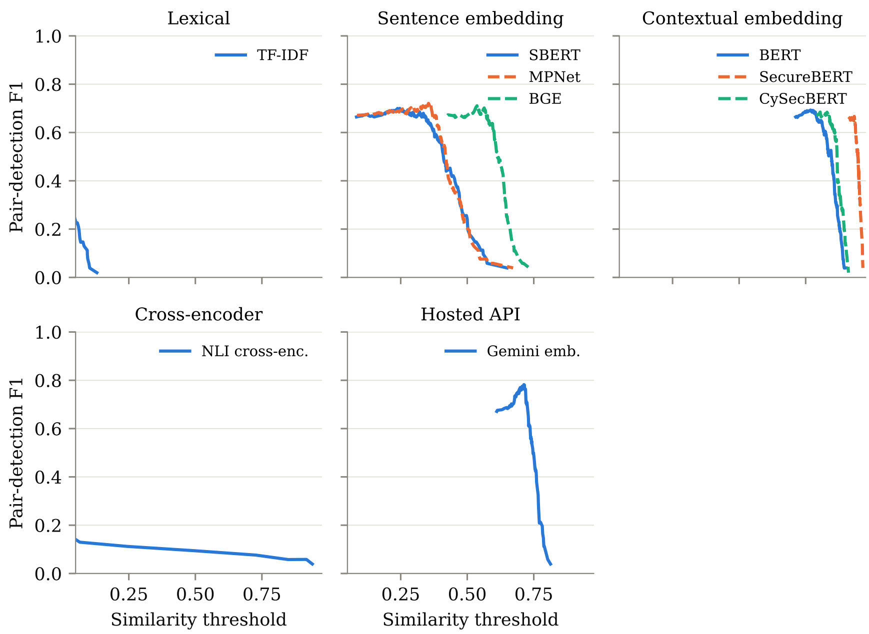

Figure 5.5: Pair-detection F1 as a function of the decision threshold, computed from the stored score matrices on the annotated scope, faceted by method family. Candidate cut points are quantiles of each method's own scores, because the methods occupy very different ranges. Eleven curves on shared axes cannot be distinguished by colour, and the families are what the comparison is about.

The methods occupy incomparable score ranges: TF–IDF tops out at 0.190 while SecureBERT never drops below 0.875. Candidate thresholds are taken as quantiles of each method's own score distribution and not from a shared absolute grid, which would hand one method a hundred usable cut points and another three. Selecting a threshold on the same pairs the score is reported on inflates it, so selection is wrapped in 5-fold cross-validation repeated 20 times, stratified on the label: the threshold is chosen on four folds and scored on the fifth, and the reported threshold is the median of the 100 selections.

Table 5.3: Detection performance when the threshold is selected by cross-validation on the reference set rather than inherited from the method's family default. Held-out F1 is the cross-validated mean; the gap to the F1 column is the remaining optimism.

Method	Threshold	Precision	Recall	Specificity	F1	Held-out F1
Gemini embeddings	0.713	**0.774**	0.790	**0.770**	**0.782**	0.756
MPNet	0.353	0.732	0.710	0.740	0.721	0.686
BGE	0.537	0.583	0.910	0.350	0.711	0.687
TF–IDF	0.001	0.572	0.910	0.320	0.703	0.689
Sentence-BERT	0.246	0.569	0.910	0.310	0.700	0.671
BERT	0.769	0.542	0.960	0.190	0.693	0.675
CySecBERT	0.804	0.521	1.000	0.080	0.685	0.668
NLI cross-encoder	0.000	0.513	0.980	0.070	0.673	0.664
SecureBERT	0.932	0.533	0.890	0.220	0.667	0.646
Label every pair related	—	*0.500*	*1.000*	*0.000*	*0.667*	—

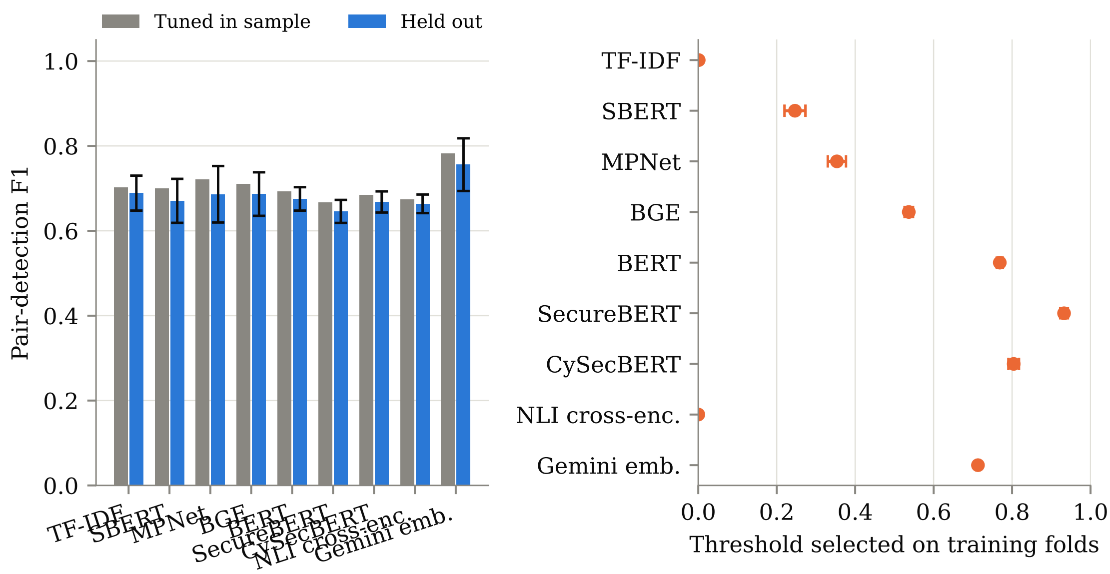

Figure 5.6: Left: pair-detection F1 with the threshold tuned on the same pairs it is scored on, against the held-out mean of 20 repeats of stratified 5-fold cross-validation (error bars, one standard deviation). Right: the threshold each set of training folds selected.

Recalibration changes the diagnosis, but far less than the shipped-threshold table might suggest. Comparisons use a prespecified contrast set fixed before the results were read: every method against the all-positive baseline, and every method against whichever method scores highest. The intervals are paired bootstrap intervals, since the methods are scored on identical pairs and the paired variance is far smaller than the marginal intervals imply.

Against the baseline, only three of the nine methods with a stored score matrix clear it by a margin whose interval excludes zero: Gemini embeddings at +0.116 [0.050, 0.188], BGE at +0.044 [0.005, 0.086] and CySecBERT at +0.018 [0.007, 0.032]. MPNet gains +0.054 but with an interval spanning zero [−0.019, 0.129], and TF–IDF, Sentence-BERT and BERT sit between +0.026 and +0.036 with intervals that also include zero. Secure-

BERT clears it by *exactly* nothing, +0.000 [−0.038, 0.036], and the NLI cross-encoder falls fractionally below at −0.004. Calibration is worth having: Gemini embeddings move from 0.667 to 0.782 and MPNet from 0.210 to 0.721. At a base rate where the trivial answer already scores 0.667, however, a recalibrated threshold buys most methods very little that survives the interval.

An interval excluding zero is not on its own a licence to read a margin as a method difference. Section 5.8 puts the reference set at $\kappa = 0.51$ on the very decision these margins are scored against, so the labels themselves carry error of an order that the smaller margins do not clear. The paired intervals for BGE (+0.044) and CySecBERT (+0.018) exclude zero because the methods are scored on identical pairs and their errors are correlated, not because either method has been shown to be better than calling everything related by an amount the annotation could resolve. Only the Gemini embeddings margin, and behind it MPNet, is large enough to survive the label noise, and the discussion below treats the ordering below that tier as undetermined.

The leader is separable this time. Measured against Gemini embeddings, every method except MPNet (−0.061, [−0.137, 0.015]) has an interval excluding zero. On this set, then, the hosted commercial embedding holds a measurable advantage over the free local models, with MPNet the only one that cannot be told apart from it. For a practitioner the relevant comparison is that MPNet reaches specificity 0.740 against the leader's 0.770 while running locally in under four seconds.

Recalibration does not rescue the domain-adapted models. SecureBERT's scores occupy a band 0.096 wide across the entire corpus, so no threshold placed anywhere in that band separates the classes, and its calibrated F1 lands precisely on the baseline. CySecBERT, adapted to the same domain from a different base model by a different group, recovers enough range to clear the baseline, but only by 0.018, and it still finishes below the plain BERT it was adapted from. Two independent attempts at cybersecurity domain adaptation both end up worse than the general-purpose model they were built from, which is a stronger statement than either could make alone.

The optimism between in-sample and held-out F1 is small everywhere (0.010 to 0.035), so a single tuned parameter over 200 points has little room to overfit. What differs across methods is the stability of the threshold itself: Gemini embeddings settle on 0.713 ± 0.004 across all 100 fits and BGE on 0.537 ± 0.011, whereas MPNet ranges over 0.353 ± 0.023. Thresholds must be calibrated per corpus and per model. Fixed literature values do not transfer, and the shipped-threshold column of Table 5.1 shows what happens when they are assumed to.

What this supports. Even recalibrated, no method here supports an automated accept/reject decision. The best F1 is 0.782 at precision 0.774, and on the far sparser full candidate space precision at the same threshold will be lower. The defensible use is the one Section 5.10 describes, ranking candidates for a human reviewer, and the choice of configuration depends on which end of the trade-off matters. Sentence-BERT at its shipped threshold surfaces 17 pairs and is right about seven times in eight; Gemini embeddings recalibrated recover 79% of the related pairs at 77% precision. Neither removes the reviewer.

5.5 Relationship classification

Detecting that two provisions relate is only half the task; compliance work needs to know *how* they relate. Table 5.4 summarises classification performance.

Every method fails at classification. The comparison that matters is against the trivial

Table 5.4: Relationship classification over all 200 annotated pairs, five-label scheme (subsumption directions merged). CIs are bootstrap 95% intervals.

Method	Accuracy	Macro-F1	Macro-F1 95% CI
TF–IDF	0.495	0.133	[0.120, 0.144]
Keyword Jaccard	0.490	**0.248**	[0.129, 0.359]
Sentence-BERT	**0.505**	0.159	[0.135, 0.186]
MPNet	**0.505**	0.159	[0.133, 0.185]
BGE	0.280	0.133	[0.101, 0.165]
BERT	0.135	0.065	[0.042, 0.089]
SecureBERT	0.015	0.006	[0.000, 0.013]
CySecBERT	0.045	0.031	[0.013, 0.050]
NLI cross-encoder	0.500	0.161	[0.126, 0.209]
Gemini embeddings	0.270	0.238	[0.117, 0.344]
Gemini LLM	0.225	0.098	[0.074, 0.124]

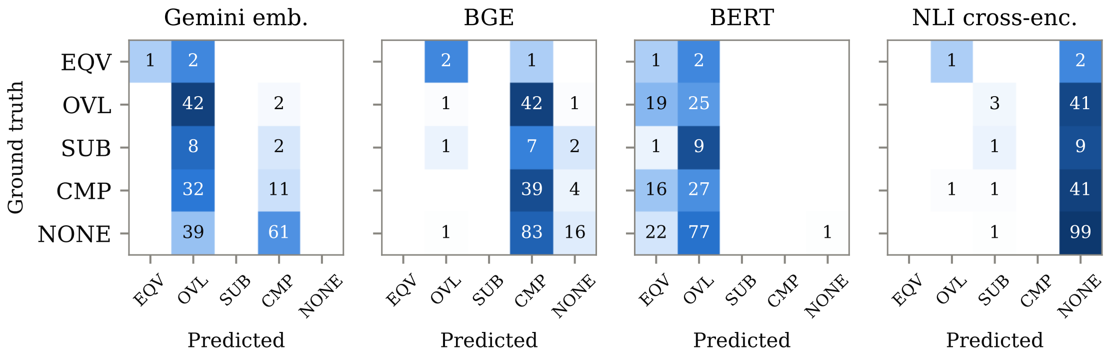

Figure 5.7: Row-normalised confusion matrices (counts printed) for the most instructive methods, one per family. EQV = equivalence, OVL = overlap, SUB = subsumption, CMP = complementarity, NONE = no relation.

classifier that assigns the majority label to everything: No relation accounts for 100 of the 200 pairs, so always predicting it yields 0.500 accuracy. The best methods match that figure without exceeding it, Sentence-BERT and MPNet at 0.505 and the cross-encoder at 0.500. The same trivial classifier scores 0.133 macro-F1, which several methods do beat, keyword Jaccard highest at 0.248 and Gemini embeddings next at 0.238, so the methods are not uniformly worthless at distributing predictions across classes. On the five-label task, however, none of the eleven methods learns the distinction the task is about.

The confusion matrices in Figure 5.7 show how each one fails, and the failures are not of one kind.

Sentence-BERT and MPNet reach the top of the accuracy column by predicting No relation for almost everything, which is the majority class on this set. Their accuracy is the majority-class strategy under another name, and their macro-F1 of 0.159 reflects it. Gemini embeddings do the opposite and funnel most predictions into Overlap, the majority class on the earlier pilot set but not on this one; accuracy drops to 0.270 while the macro-F1 is the second best of the eleven. That method also reaches Equivalence at precision 1.00 and recall 0.33, recovering one of the three equivalence pairs. BERT and the two cybersecurity models simply collapse: SecureBERT's uniform Equivalence output produces 0.015 accuracy, exactly the three equivalence pairs in the reference set, and CySecBERT reaches 0.045.

The two methods that are not similarity thresholds behave differently again. The Gemini LLM predicts labels from provision content instead of a score band, and it does recognise complementarity, at recall 0.84. It also heavily overuses that label, assigning a relation to 91 of the 100 annotated negatives and reaching No relation with recall 0.09. The NLI cross-encoder is the only method that predicts Subsumption at all, for the reason given in Section 3.5.4: over the full corpus it emits 123 subsumption labels where every other method emits none. Its macro-F1 of 0.161 is nonetheless unremarkable, because the method is extremely conservative. It flags 142 of 4,968 pairs, and so reaches the directional label on too few pairs to score well. This is the one gap in the taxonomy that a change of architecture closes and a better-calibrated threshold does not.

The scarcity documented in Section 5.1 bounds what any of this can establish. Equivalence is supported by three pairs and Subsumption by ten, so the per-class scores for those two rows move by large amounts on a single prediction and the macro-F1 that averages them inherits that instability. The failure is measured on the classes where the reference set has real support, Overlap, Complementarity and No relation, and it is on the overlap/complementarity boundary that both the methods and the annotators disagree most. The validation study below quantifies that.

5.6 What supervision buys

Every method in Table 5.4 is zero-shot: it turns a similarity score into a relationship type through the fixed bands of Section 3.5.6, and Section 3.5.4 already established that a symmetric scalar cannot express what the taxonomy asks for. The failure reported above is therefore a failure of that framing, and it leaves a question open that the project has the material to answer. Of the 1,027 annotated pairs, only 200 are scored on. The remaining 827 are available as training data.

This section reports what a classifier trained on them achieves. It is a control and not a twelfth method: no model is trained, downloaded or called: the features are the scores the eleven methods have already produced, so the question it answers is whether those representations carry typed information that the fixed bands fail to extract. Two classifiers are fitted, a logistic regression over standardised features and a gradient-boosted tree ensemble, both with balanced class weights, and both are evaluated on the same 200 pairs as everything else.

The split determines what can be claimed. The 827 training pairs hold only NO RELATION (601), OVERLAP (130) and COMPLEMENTARITY (96): every EQUIVALENCE and SUBSUMPTION pair in the corpus is in the evaluation set, a consequence of the composition rule of Section 3.4.3 drawing those classes at their full available count. The trained classifier therefore cannot predict either rare class by construction, and typed results are reported twice: over all 200 pairs, where the 13 rare-class pairs are counted as errors and the figure is comparable with Table 5.4, and over the 187 pairs whose classes the training data covers. The training pool is also 73% negative against the evaluation set's 50%, so the classifiers meet a positive rate they were not fitted for, while the zero-shot thresholds of Table 5.3 were calibrated on the evaluation set itself. The comparison below is if anything unfavourable to supervision.

Table 5.5: A classifier trained on the 827 judged pairs held out of the reference set, using the eleven methods' stored scores as features. Typed metrics are given over all 200 pairs, with the 13 rare-class pairs counted as errors, and over the 187 pairs whose classes the training data covers. The zero-shot row is the best value in each column taken separately from Tables 5.3 and 5.4, so it is a per-column ceiling rather than any single method.

	Detection		Typed classification		
Model	Precision	F1	Accuracy	Macro-F1	Macro-F1 (187)
Logistic regression	0.778	**0.737**	**0.580**	**0.321**	**0.555**
Gradient boosting	**0.848**	0.629	0.550	0.250	0.433
Best zero-shot method	*0.774*	*0.782*	*0.505*	*0.248*	—
Trivial baseline	*0.500*	*0.667*	*0.500*	*0.133*	*0.232*

The result splits cleanly by task, and the split is the point.

On detection, supervision does not help. The logistic regression reaches F1 0.737 [0.663, 0.800], below the 0.782 of calibrated Gemini embeddings, and its margin over the all-positive baseline is +0.070 with a paired interval spanning zero [−0.007, 0.146]. Eleven scores combined by a trained model do not beat the best single score thresholded well. Whatever limits detection here is not the crudeness of a single cut point.

On typing, supervision does help, and by enough to change the claim Section 5.5 is entitled to make. The logistic regression reaches 0.580 accuracy where every zero-shot method matched the majority-class strategy at 0.505 or below, and macro-F1 0.321 [0.279, 0.362] against the trivial 0.133 and the best zero-shot 0.248. Restricted to the three classes the training data covers, it reaches macro-F1 0.555 [0.477, 0.624] against a majority-class 0.232. The typed information is present in the same scores the eleven methods produce; the fixed similarity bands simply do not extract it. What Section 5.5 measures is that zero-shot typing through score bands fails, which is a narrower and more defensible claim than that the typed task is beyond reach.

That said, the trained model does not solve the distinction the taxonomy turns on. Its errors concentrate exactly where the annotators' do: of the 43 COMPLEMENTARITY pairs it labels 17 as OVERLAP and only 13 correctly, and it assigns a relation to 25 of the 100 annotated negatives. The overlap/complementarity boundary, which Section 5.8 shows the annotation itself resolves at $\kappa = 0.40$, is not a boundary supervision learns from 827 examples either.

The fitted coefficients are informative for RQ2 in a way the zero-shot tables are not.

The largest standardised weights fall on Gemini embeddings (+0.74) and Sentence-BERT (+0.52), with BERT entering negatively (−0.33): the trained model uses the sentence-embedding and hosted representations and discounts the mean-pooled contextual ones, which is the same ordering the calibrated results produce and is reached here without any threshold being chosen.

Two limits apply. The 827 training pairs and the 200 evaluation pairs come from the same three annotation rounds and the same 141 provisions, so this measures generalisation to new pairs and not to new standards. And the classifier inherits every property of the features: it cannot recover a distinction that none of the eleven underlying scores encodes, which is why it reaches SUBSUMPTION not at all.

5.7 Coverage and gaps

The preceding sections measure how often a method is right. This one asks what the method actually says about the two standards, which is the question a compliance practitioner brings to a mapping exercise: which provisions of ETSI EN 303 645 have a counterpart in ETSI EN 304 223, and which have none. The manual study closest to this one [7] reports a coverage statement and an explicit list of unmapped requirements, so producing the same two artefacts automatically is the fairest way to show what the automation delivers.

Everything in this section is a prediction from a single method, the best-scoring one at its calibrated operating point. Its measured precision is 0.774 [0.691, 0.851] on the reference set, which by the argument of Section 3.4.3 is an upper bound on its precision across all 4,968 candidates. What the numbers below establish is the shape of the predicted mapping, not its accuracy.

At its calibrated threshold the method flags 1,018 of the 4,968 candidate pairs, 20.5% of the space. That is a high rate for a corpus whose related pairs are plainly a minority, and it follows from fitting a threshold where positives are half the data: the operating point is tuned for a base rate the corpus does not share, so at corpus scale it over-predicts.

Figure 5.8 shows where the predicted relations concentrate. The density is uneven in a way that follows the structure of the two documents rather than the behaviour of the model.

The gaps are the part a similarity threshold can state cleanly. Seven of the 69 ETSI EN 303 645 provisions and three of the 72 ETSI EN 304 223 provisions receive no predicted counterpart at all; Table 5.6 lists them. A gap here is a weaker claim than a gap in a manual study: it means the method found nothing, not that nothing is there, and at a recall below one the two are different statements. The list is still useful, because checking a false gap costs an analyst ten provisions and missing a real one costs more.

The full predicted mapping, all 1,018 pairs with their scores, is given in Appendix D. Both the table above and that appendix are generated directly from the evaluation outputs, so neither can drift from the numbers reported here.

5.8 Ground-truth validation

A reference set is only as good as the consistency of the judgements behind it. The pilot pairs were mixed unmarked into the annotation stream of the later round, so re-labelling them was blind: at the time of judging, there was no way to tell a pilot pair from a newly screened one. Comparing the two label sets afterwards gives an agreement figure against annotations that two authors reviewed months earlier.

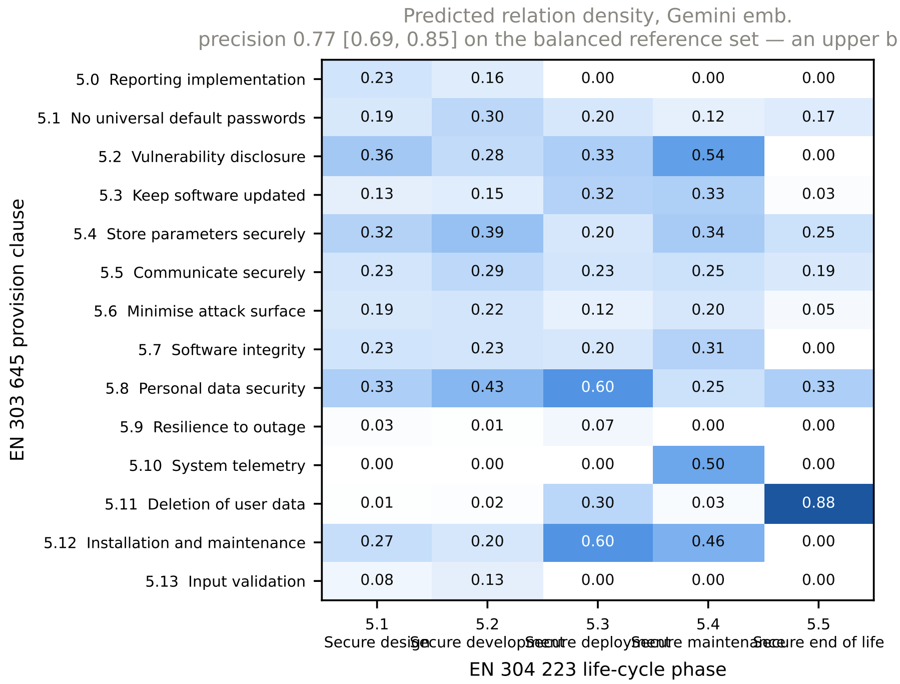

Figure 5.8: Predicted relation density by clause. Each cell is the share of provision pairs in that clause block flagged as related by Gemini embeddings at the calibrated threshold of 0.713. These are predictions, not verified relations, and the method's precision of 0.774 [0.691, 0.851] was measured on a balanced set and is an upper bound at corpus scale.

Table 5.6: Provisions with no predicted counterpart in the other standard, at the calibrated operating point. Absence of a prediction is not evidence of absence of a relation; the method's recall at this threshold is below one.

Standard	Provision	Requirement
EN 303 645	5.1-2A	Passwords should not be used for machine-to-machine authentication. Using hum...
	5.3-15A	For devices that cannot have their software updated, the consumer IoT device...
	5.5-3	Cryptographic algorithms and primitives should be replaceable.
	5.5-4	Access to consumer IoT device functionality via a network interface in the in...
	5.6-4B	Debug interfaces that are physical ports should be physically protected by th...
	5.6-6	Code should be minimized to the functionality necessary for the consumer IoT...
	5.9-3	The consumer IoT device should connect to networks in an expected, operationa...
EN 304 223	5.1.1-1.1	AI security training shall be tailored to the specific roles and responsibili...
	5.1.2-1.1	Where the Data Custodian is part of a Developer's organization, they shall be...
	5.2.5-2	System Operators shall conduct testing prior to the system being deployed wit...

Table 5.7: Agreement of the second annotation round with the earlier author-reviewed pilot labels, against the same statistic computed for the evaluated LLM on two runs over identical input.

		Six labels		Related vs. not	
Comparison	n	Agreement	κ	Agreement	κ
Second round vs. reviewed pilot	107	0.61	0.40	0.82	0.51
Evaluated LLM vs. itself, identical input	4 968	0.69	0.02	0.82	0.04

Agreement with the pilot reaches $\kappa = 0.51$ on the detection decision and $\kappa = 0.40$ on the six-label scheme, moderate and fair respectively on the Landis–Koch scale [36]. A reference set at $\kappa = 0.51$ on its central decision is usable for comparing methods against each other, since every method is scored against the same labels, but it is not precise enough to certify an absolute performance level. The rankings in Section 5.4 should be read with that ceiling in mind.

Disagreements are not scattered across the label space. They cluster on the overlap/complementarity boundary, which Section 5.11 identifies as the least determinate distinction in the taxonomy, and on the general-process provisions covered by the codebook amendment recorded during annotation. Recorded confidence tracks them: pairs marked confidence 3 agree with the pilot 77% of the time ($n = 51$) against 46% for confidence 2 ($n = 56$). The comparison rests on the confidence recorded in the second round, which used only levels 2 and 3, so this is a two-level contrast and not a calibration curve. Level 1 was available throughout and was used sparingly in the pilot, five times across its 107 pairs, three of which reach the reference set. It does show that the field carries information.

The second row of Table 5.7 is the more instructive one. The evaluated LLM was run twice at temperature 0 over byte-identical prompts. The two runs agree on 69% of pairs, *higher* raw agreement than the human comparison, yet reach $\kappa = 0.02$. The model assigns COMPLEMENTARITY to 81% of all pairs, so two runs coincide frequently without either run carrying information. Raw agreement rewards a degenerate marginal; κ does not. The reliability of an annotation source cannot be read off its agreement percentage, and the LLM classifier's headline scores should not be taken at face value.

5.9 Efficiency and interpretability

Every local method runs in seconds or minutes on commodity hardware (Table 5.8). By comparison, the three annotation rounds behind the reference set occupied several working weeks even with LLM assistance. The cost structure separates the two architectures cleanly. A bi-encoder embeds each provision once, so the work is linear in the 141 provisions and the 4,968 similarities are a single matrix product, which is why every embedding stage finishes in under six seconds. A cross-encoder must run the model once per pair per direction, so its work is quadratic in the provisions: 9,936 forward passes and 327 seconds, roughly sixty times the cost of the slowest bi-encoder for the same corpus. On this pair of standards that is affordable; on a pair ten times larger it would not be.

Interpretability differs more than runtime. The keyword baseline can show *which* terms matched, embedding methods offer only an opaque score, the cross-encoder additionally reports which direction the entailment ran, and the LLM can produce a textual rationale on request, though its rationale quality was not separately evaluated here.

Table 5.8: Wall-clock runtime per stage on an 8 GB Apple silicon laptop, including model loading but not the one-off model download (about 2 GB in total). Embedding stages use the Metal backend. Gemini stages run through the batch API; their end-to-end time is dominated by queueing (minutes to hours) rather than computation.

Stage	Runtime
Provision extraction (both PDFs)	0.9 s
TF–IDF + keyword mapping	3.5 s
Sentence-BERT mapping	1.8 s
MPNet mapping	3.9 s
BGE mapping	3.9 s
BERT mapping	4.2 s
SecureBERT mapping	5.7 s
CySecBERT mapping	3.9 s
NLI cross-encoder (9,936 forward passes)	327 s
Gemini embeddings	API batch
Gemini LLM classification (4,968 prompts)	API batch
Evaluation	1.4 s
Calibration and bootstrap analysis	58 s

5.10 Discussion

RQ1 — feasibility and limits. Automated methods can *shortlist* related provisions, and calibration makes the difference between a usable ranking and a constant function. At their shipped thresholds four of the eleven methods reproduce the all-positive baseline exactly, while a cross-validated threshold lifts the best of them to F1 0.782 at precision 0.774. That margin over the trivial answer is real but modest, +0.116 on a baseline of 0.667, and it is measured where positives are half the data. Identifying the *type* of relationship is not solved by any of the eleven. The best classification accuracy, 0.505, is the majority-class strategy under another name. The reason is structural: the overlap/complementarity boundary requires judging *objectives* rather than surface meaning, and a scalar similarity read through fixed bands collapses exactly the distinctions the taxonomy is designed to capture.

The supervised control of Section 5.6 locates that failure more precisely, and the location is not where the zero-shot tables suggest. Trained on the 827 judged pairs held out of the reference set, using nothing but the same eleven methods' stored scores, a logistic regression reaches 0.580 accuracy and macro-F1 0.321 where the best zero-shot method reaches 0.505 and 0.248. The typed signal is present in those representations and the fixed bands do not extract it, so what Table 5.4 demonstrates is the failure of zero-shot typing through similarity thresholds rather than the intractability of typing as such. The qualification matters in both directions: the same trained model still labels more COMPLEMENTARITY pairs OVERLAP than correctly, so the boundary the taxonomy turns on survives supervision as well, and none of this reaches the accuracy compliance documentation would require.

One of those limits turns out to be architectural. Symmetric similarity cannot express directional subsumption, but the NLI cross-encoder, which asks a directional question by construction, predicts 123 subsumption relations where the other ten methods predict none. Its recall is too low for it to be useful on its own, so the practical conclusion is unchanged. The distinction the taxonomy needs is reachable, though not by the family of methods that dominates this literature.

A finding that emerged from the annotation rather than the models belongs here too. The two rarest relations are rare in the corpus and not only in the reference set: 13 pairs of the 1,027 judged, and none among the 300 highest-ranked unjudged pairs searched specifically to find more. For this pair of standards the typed taxonomy is effectively a three-class problem, and the two classes that motivated the directional machinery barely occur.

RQ2 — method comparison. No method family wins outright. They fail differently, and with three models in each of the two embedding families the failures separate by family and not by model. Lexical baselines are precise but nearly blind at their shipped thresholds (recall ≤ 0.06), though TF–IDF recalibrated reaches F1 0.703, which says more about how weak the bar is at this base rate than about TF–IDF. Contextual embeddings saturate: anisotropy makes their cosine scores unusable without calibration, and all three sit at the bottom of the calibrated ranking. Domain pretraining makes this worse, and that now rests on two independent models, SecureBERT and CySecBERT, built by different groups from different base models, both of which finish below the plain BERT they were adapted from. The intuition that cybersecurity vocabulary is the missing ingredient is contradicted twice.

Gemini embeddings top the calibrated table at F1 0.782 and the paired bootstrap separates them from every method except MPNet (-0.061, $[-0.137, 0.015]$). The practical answer to whether the paid model is needed comes out qualified. MPNet is the only free local model that cannot be told apart from it. It runs in under four seconds with no per-query cost and reproduces bit-identically across runs, whereas the Gemini embeddings depend on a hosted model that can change under the same name, and the LLM replicate of Section 5.8 shows what hosted non-determinism costs. Calibrated MPNet for ranking, plus human labelling of the shortlist, is the defensible configuration; the paid model earns its place where the marginal F1 it may carry is worth the loss of reproducibility. The cross-encoder buys direction at roughly sixty times the compute and a recall too low to exploit it. The prompted LLM has non-trivial recall on complementarity but over-predicts it massively and rejects almost no negative.

RQ3 — automated versus human mapping. The standard of comparison is the reference set of Section 3.4, which is LLM-drafted, cross-verified and author-reviewed rather than expert-built, so what follows compares automation against that reference and not against a panel of domain experts. Against it, automation is dramatically faster (seconds versus weeks) and consistent, but its label agreement remains far below what compliance documentation requires: the best typed accuracy matches the majority-class strategy. These findings support the same conclusion Bar-Haim et al. [13] reached for control mapping: automated compliance mapping is a decision-support technology. Concretely, the evidence here supports a workflow where an embedding model ranks all 4,968 pairs, a human reviews the top of the ranking, and the relationship label is always assigned by the human.

5.11 Threats to validity

Construct validity. The six-label taxonomy compresses a continuum of relationships between requirements into discrete classes; the overlap/complementarity boundary in particular is partly a judgement call. Section 5.8 puts a number on the cost: agreement on the binary related/unrelated question reaches $\kappa = 0.51$, and resolving *which* of the six relations holds falls to $\kappa = 0.40$, so the typed labels are the noisier construct and the classification results in Section 5.5 are measured against a correspondingly noisier target.

Internal validity. Provision text from ETSI EN 303 645 retains the unlabelled explanatory paragraphs the standard prints beneath its requirements, as Section 3.2 sets out, and those paragraphs are 53% of that standard's extracted words. They are written in general security vocabulary, so two provisions whose requirements are unrelated can still share a substantial amount of wording through their explanations alone. Every similarity score in this chapter is computed over text of that composition, so the compression documented in Section 5.3 could in principle be the input rather than the models. The ablation reported in Section 5.3 measures that contribution instead of assuming it: re-scoring the nine local methods on normative sentences alone, which removes 55% of ETSI EN 303 645's words, moves no method's calibrated detection F1 by as much as 0.03 and leaves SecureBERT confined to a band 0.119 wide and scoring exactly the all-positive baseline. The input contributes something to the crowding, since the compressed models' score ranges do widen slightly, but not the effect. This threat is measured and largely discharged for the local methods. It remains open for the two hosted ones, which are archived as stored outputs and could not be re-scored on the same footing — and one of those leads the comparison, so the qualification is not a small one.

The reference set was drafted with LLM assistance, which risks favouring LLM-family methods. Four things bound the risk. Neither annotation model belongs to the evaluated family: the labels were drafted by Fable 5 (Anthropic) and, in the screened and targeted rounds, verified by GPT-5.6 Sol (OpenAI), while the evaluated methods are Gemini. In those rounds the two annotation models are independent of each other, so a label reflects agreement across developers rather than one model's judgement. 37 of the 200 pairs come from the author-reviewed pilot. And the evaluated LLM classifier performed worst of the non-degenerate methods on the typed task, so the circularity, if any, did not manifest as an advantage. The two warrants do not overlap, which is the honest form of this limitation: the 37 pilot pairs were read individually by both authors but never cross-verified, and the remaining 163 were cross-verified but not reviewed by an author pair by pair, carrying the two models' agreement plus author adjudication where the models diverged. The reliability of that second group rests further on the blind comparison with the reviewed pilot reported in Section 5.8.

External validity. Two limits bound generalisation. The first is the balanced composition of the reference set: as Section 3.4.3 sets out, every figure in Sections 5.2–5.5 is an upper bound on what the same threshold achieves at corpus scale, and how much lower is not quantified here. The second is scope. The results belong to this pair of standards and these model versions, and the specific numbers should not be transferred to another pair without re-measurement.

Statistical conclusion validity. Five limits constrain how far the numbers can be pushed. First, the rare classes are rare: with three EQUIVALENCE and ten SUBSUMPTION pairs, per-class figures for those two rows move substantially on a single prediction, and the macro-F1 that averages them inherits that instability. The targeted search reported in Section 5.1 establishes that this cannot be fixed by annotating more pairs from the same corpus. The same scarcity reaches the intervals: a bootstrap resample of 200 pairs omits all three EQUIVALENCE pairs about one time in twenty, and that class then contributes zero to the macro average, so the macro-F1 intervals in Table 5.4 are biased slightly low and are wider than the underlying estimate warrants. The support for those two rows is itself label-dependent, as Section 3.4.3 records: twelve of the thirteen rare-class pairs carry pilot labels, and under the blind re-annotation of the same pairs the pool would hold two EQUIVALENCE and four SUBSUMPTION rather than three and ten. Neither reading changes the scarcity finding, but the per-class rows for those two classes rest on the pilot round specifically and not on the reference set as a whole. Second, the operating points in Table 5.3 are selected by cross-validation on the same set they are scored on. The residual optimism is measured rather than assumed, at between 0.010 and 0.035 across methods, and is small relative to the differences reported, but the point estimates remain the optimistic end of their range. Third, the paired bootstrap holds each method's calibrated threshold fixed across the resamples rather than re-selecting it within each one, so those intervals carry the sampling variance of the metric at a fixed operating point and not the variance of the selection procedure that chose it. They are consequently narrower than a fully nested resampling would produce. Fourth, the paired comparisons follow a prespecified contrast set, each method against the baseline and against the leader, instead of all pairs, because fifty-five uncorrected intervals would guarantee false findings among them. The leader is nonetheless the sample maximum, so differences measured against it are biased in its favour and are read as descriptive rather than as a test that it is best. Fifth, every bootstrap here resamples the 200 pairs independently, but the pairs are not independent: they are drawn from 141 provisions, so the pairs sharing a source or a target provision move together, and a single ambiguously worded provision perturbs every pair it appears in. Resampling pairs rather than provisions therefore treats correlated observations as independent and understates the variance. A cluster bootstrap over provisions would widen every interval in this chapter; the direction of the correction is known, its size is not.

6 Conclusion

This thesis asked how far the mapping of security requirements between ETSI EN 303 645 and ETSI EN 304 223 can be automated with current NLP and LLM techniques. A five-stage pipeline was built that extracts 141 normative provisions from the two standards, maps them with eleven methods spanning lexical, sentence-embedding, contextual-embedding, cross-encoder and hosted API approaches, and evaluates all of them against a 200-pair reference set assembled from three rounds of LLM-assisted annotation totalling 1,027 judged pairs: every label drafted by one frontier model, those of the later two rounds verified by a second from an unrelated developer, with the pilot round read pair by pair by both authors and the later rounds adjudicated by them wherever the two models diverged.

The answer the evidence supports is a split one. Finding *that* two provisions relate is partly feasible; determining *how* they relate is not.

The detection result turns on calibration. The reference set was built to be half related and half not, so the classifier that calls every pair related scores $F_1 = 0.667$ on it, and that figure is the bar every method has to clear. At the thresholds they ship with, four of the eleven methods do not clear it at all. SecureBERT, CySecBERT, Gemini embeddings and, to within one pair, BERT mark *every* annotated pair as related and reproduce the trivial score exactly. Re-selecting each threshold by cross-validation changes that picture, though less than the collapse might suggest. Gemini embeddings reach F1 0.782 at precision 0.774, a margin of +0.116 [0.050, 0.188] over the baseline; BGE and CySecBERT clear it by smaller margins whose intervals also exclude zero, though at a reference set measuring $\kappa = 0.51$ on that decision those two margins are smaller than the label noise and are not read as method differences; and the remaining methods gain something whose interval does not. Calibration is therefore worth having. The practical lesson is stronger than a warning about optimistic benchmarks: a threshold fixed without knowing where a model's scores fall can turn a working model into a constant function. It does not, however, turn a modest margin into a large one.

Determining how two provisions relate is not solved by any of the eleven. The best typed accuracy, 0.505, is what predicting the majority class for every pair achieves, and no evaluated method exceeds it. The reason is the framing rather than the representations: a control trained on the 827 judged pairs held out of the reference set, using only the scores those same eleven methods produce, reaches 0.580 accuracy and macro-F1 0.321 against a trivial 0.500 and 0.133. The typed signal is in the scores and the fixed similarity bands do not extract it. What that control does not do is resolve the overlap/complementarity boundary, where it errs as the annotators do, so typed mapping remains well short of what compliance documentation requires. No similarity-based method can represent directional subsumption even in principle, and the cross-encoder added here to test that limit confirms both halves of it: reading each pair in both directions does recover subsumption, producing 123 such labels where the other ten methods produce none, but at a recall too low to be useful and at roughly sixty times the compute.

Two failure mechanisms recur across the experiments: the anisotropy of general-purpose embedding spaces, which drives BERT-family models into degenerate "everything is similar" behaviour, and the mismatch between scalar similarity and a typed relationship taxonomy. Cybersecurity-specific pretraining did not help in either of the two independent attempts tested (SecureBERT, CySecBERT), while task-specific training for sentence comparison did help in all three (Sentence-BERT, MPNet, BGE). What the model was trained to do predicts its behaviour here; what it was trained on does not.

A third finding came from the annotation rather than the models. Equivalence and subsumption are close to absent from the corpus: 13 pairs among the 1,027 judged, and none at all among the 300 highest-ranked unjudged pairs annotated specifically to find more. A device-level standard and a lifecycle-level standard produce overlapping and complementary obligations in quantity, and identical or strictly-containing ones almost never. For this pair of standards the six-label taxonomy is effectively a three-label one.

For practice, the results position automated compliance mapping as decision support: rank candidates automatically, let humans assign the relationship labels. Even that reduced role is valuable to a compliance team, since the ranking step compresses 4,968 candidate pairs into a short review list in under a minute of computation.

Critical discussion

Three choices shaped this project and deserve scrutiny. First, the reference set was annotated with LLM assistance; this was a pragmatic necessity at 15 credits, it is disclosed openly, and the design (two annotation models from separate developers, neither sharing a family with the evaluated ones, a written codebook fixed in advance, author adjudication of every model disagreement, and a blind comparison against the author-reviewed pilot) bounds the risk. What it does not remove is the ceiling that comparison reveals: $\kappa = 0.51$ on the detection decision means the reference is good enough to rank methods against a common target but not to certify an absolute accuracy. A fully expert-built reference would be stronger, and the survey in Appendix B is the direct path to one.

Second, and most consequentially, the reference set was built to a positive rate of one half. That choice succeeded in what it was for: the degeneracy of four methods is visible directly in Table 5.1 instead of being hidden behind a high number. One half is nonetheless a design parameter and the corpus is far sparser, so every performance figure in this thesis stands as an upper bound on what the same operating point achieves across all 4,968 candidates (Section 3.4.3). The size of that gap cannot be measured from this design. The coverage analysis of Section 5.7 should be read accordingly, as the shape of the predicted mapping and not its accuracy at corpus scale.

Third, reporting two operating points per method rather than one doubles the tables but is the only way to separate what a model can represent from where its threshold happens to sit.

Generalisation of the results

The specific numbers belong to this standard pair, this reference set and these model versions. The structural findings travel further: anisotropy of mean-pooled contextual embeddings is a documented model property, not a dataset accident; the inability of symmetric scores to express asymmetric relationships holds for any standard pair; and the observation that detection is calibration-limited while typed classification fails outright should recur wherever two related security frameworks are mapped. The pipeline itself is standard-agnostic, and any pair of documents with identifiable normative clauses can be substituted.

Future work

Four directions follow naturally, and the first is a direct repair of this design's principal limitation.

First, measure performance at the corpus base rate. Every figure here is conditioned on a reference set built to be half positive, and the gap between that and the sparse reality of 4,968 candidate pairs is currently unquantified. Drawing a sample with known inclusion probabilities over the full space and estimating precision, recall and prevalence from it would convert the upper bounds reported here into estimates, and would put a number on how far the calibrated thresholds of Table 5.3 drift when applied where positives are rare. This is the single change that would most improve the strength of the claims.

Second, execute the expert validation survey designed in Appendix B to put the ground truth on a multi-rater footing and to calibrate the human agreement ceiling.

Third, train the typed task properly. The control in Section 5.6 shows that supervision over the stored scores alone lifts typed accuracy above every zero-shot method, which makes end-to-end training the obvious extension rather than a speculative one: a cross-encoder fine-tuned on requirement pairs, rather than borrowed from general natural language inference, would train the representation and the decision together, and entailment scoring is in any case the only approach here that recovers subsumption direction. The scarcity finding tempers this: with thirteen rare-class pairs in the whole corpus, the signal for those two classes would have to come from other standard pairs.

Fourth, revisit the prompt given to the LLM classifier. Its overuse of complementarity tracks the wording of the label definition rather than the difficulty of the pairs, since "related but different concerns that work together" admits almost any two security requirements. The ceiling measured here may therefore be the definition's rather than the model's.

Beyond these, ranking metrics such as precision at k and mean average precision would decouple the evaluation from the positive rate altogether, since they measure the shortlisting behaviour the pipeline is actually fit for; the machinery is implemented but the judged fraction of each ranking is still so thin that precision among judged candidates is 1.000 for every method, meaning the numbers describe judging coverage rather than ranking quality.

Bibliography

[1] M. M. Aslam, A. Tufail, H. Gul, *et al.*, "Artificial intelligence for secure and sustainable industrial control systems – a survey of challenges and solutions," *Artificial Intelligence Review*, vol. 58, p. 349, 2025. `https://doi.org/10.1007/s10462-025-11320-9`.

[2] B. Alotaibi, "A survey on industrial Internet of Things security: Requirements, attacks, AI-based solutions, and edge computing opportunities," *Sensors*, vol. 23, no. 17, p. 7470, 2023. `https://doi.org/10.3390/s23177470`.

[3] H. Meziane and N. Ouerdi, "A survey on performance evaluation of artificial intelligence algorithms for improving IoT security systems," *Scientific Reports*, vol. 13, p. 21255, 2023. `https://doi.org/10.1038/s41598-023-46640-9`.

[4] ETSI, "ETSI EN 303 645 V3.1.3: Cyber security for consumer internet of things: Baseline requirements." European Telecommunications Standards Institute, 2024.

[5] ETSI, "ETSI EN 304 223 V2.1.1: Securing artificial intelligence (SAI); baseline cyber security requirements." European Telecommunications Standards Institute, 2025.

[6] O. Kosenkov, P. Elahidoost, T. Gorschek, J. Fischbach, D. Mendez, M. Unterkalmsteiner, D. Fucci, and R. Mohanani, "Systematic mapping study on requirements engineering for regulatory compliance of software systems," *Information and Software Technology*, vol. 178, p. 107622, 2025. `https://doi.org/10.1016/j.infsof.2024.107622`.

[7] F. Djebbar and K. Nordström, "A comparative analysis of industrial cybersecurity standards," *IEEE Access*, vol. 11, pp. 85315–85332, 2023. `https://doi.org/10.1109/access.2023.3303205`.

[8] D. Canavese, A. Ferreira, L. Kamm, and A. Q. Rodriguez, "Uncovering challenges of cybersecurity cross-regulation in EU legislation," *Information and Software Technology*, vol. 193, p. 108033, 2026. `https://doi.org/10.1016/j.infsof.2026.108033`.

[9] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, (Minneapolis, Minnesota), pp. 4171–4186, Association for Computational Linguistics, 2019. `https://doi.org/10.18653/v1/N19-1423`.

[10] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, (Hong Kong, China), pp. 3982–3992, Association for Computational Linguistics, 2019. `https://doi.org/10.18653/v1/D19-1410`.

[11] D. Rodriguez, I. Yang, J. M. Del Alamo, and N. Sadeh, "Large language models: a new approach for privacy policy analysis at scale," *Computing*, vol. 106, no. 12, pp. 3879–3903, 2024. https://doi.org/10.1007/s00607-024-01331-9.

[12] O. A. Cejas, M. I. Azeem, S. Abualhaija, and L. C. Briand, "NLP-based automated compliance checking of data processing agreements against GDPR," *IEEE Transactions on Software Engineering*, vol. 49, no. 9, pp. 4282–4303, 2023. https://doi.org/10.1109/tse.2023.3288901.

[13] R. Bar-Haim, L. Eden, Y. Kantor, V. Agarwal, M. Devereux, N. Gupta, A. Kumar, M. Orbach, and M. Zan, "Towards automated assessment of organizational cybersecurity posture in cloud," in *Proceedings of the 6th Joint International Conference on Data Science & Management of Data (CODS-COMAD '23)*, (New York, NY, USA), pp. 167–175, Association for Computing Machinery, 2023. https://doi.org/10.1145/3570991.3571008.

[14] S. Sicari, A. Rizzardi, L. Grieco, and A. Coen-Porisini, "Security, privacy and trust in Internet of Things: The road ahead," *Computer Networks*, vol. 76, pp. 146–164, 2015. https://doi.org/10.1016/j.comnet.2014.11.008.

[15] G. Vardakis, G. Hatzivasilis, E. Koutsaki, and N. Papadakis, "Review of smart-home security using the Internet of Things," *Electronics*, vol. 13, no. 16, 2024. https://doi.org/10.3390/electronics13163343.

[16] D. Mocrii, Y. Chen, and P. Musilek, "IoT-based smart homes: A review of system architecture, software, communications, privacy and security," *Internet of Things*, vol. 1–2, pp. 81–98, 2018. https://doi.org/10.1016/j.iot.2018.08.009.

[17] H. Chen and M. A. Babar, "Security for machine learning-based software systems: A survey of threats, practices, and challenges," *ACM Computing Surveys*, vol. 56, 2024. https://doi.org/10.1145/3638531.

[18] X. He, G. Xu, X. Han, Q. Wang, L. Zhao, C. Shen, C. Lin, Z. Zhao, Q. Li, L. Yang, *et al.*, "Artificial intelligence security and privacy: a survey," *Science China Information Sciences*, vol. 68, no. 8, p. 181101, 2025. https://doi.org/10.1007/s11432-025-4388-5.

[19] A. Mussmann, M. Brunner, and R. Breu, *Mapping the State of Security Standards Mappings*, pp. 1309–1324. GITO Verlag, 2020. https://doi.org/10.30844/wi_2020_l4-mussmann.

[20] C. Negri-Ribalta, M. Lombard-Platet, and C. Salinesi, "Understanding the GDPR from a requirements engineering perspective — a systematic mapping study on regulatory data protection requirements," *Requirements Engineering*, vol. 29, no. 4, pp. 523–549, 2024. https://doi.org/10.1007/s00766-024-00423-4.

[21] J. P. Castellanos Ardila, B. Gallina, and F. Ul Muram, "Compliance checking of software processes: A systematic literature review," *Journal of Software: Evolution and Process*, vol. 34, no. 5, p. e2440, 2022. https://doi.org/10.1002/smr.2440.

[22] A. Sleimi, N. Sannier, M. Sabetzadeh, L. Briand, M. Ceci, and J. Dann, "An automated framework for the extraction of semantic legal metadata from legal texts," *Empirical Software Engineering*, vol. 26, 2021. https://doi.org/10.1007/s10664-020-09933-5.

[23] A. Mahmoud and N. Niu, "On the role of semantics in automated requirements tracing," *Requirements Engineering*, vol. 20, no. 3, pp. 281–300, 2015. https://doi.org/10.1007/s00766-013-0199-y.

[24] D. Rajpathak, P. M. Peranandam, and S. Ramesh, "Automatic development of requirement linking matrix based on semantic similarity for robust software development," *Journal of Systems and Software*, vol. 186, p. 111211, 2022. https://doi.org/10.1016/j.jss.2021.111211.

[25] L. Zhao, W. Alhoshan, A. Ferrari, K. J. Letsholo, M. A. Ajagbe, E.-V. Chioasca, and R. T. Batista-Navarro, "Natural language processing for requirements engineering: A systematic mapping study," *ACM Computing Surveys*, vol. 54, no. 3, pp. 1–41, 2021. https://doi.org/10.1145/3444689.

[26] R. Sonbol, G. Rebdawi, and N. Ghneim, "The use of NLP-based text representation techniques to support requirement engineering tasks: A systematic mapping review," *IEEE Access*, vol. 10, pp. 62811–62830, 2022. https://doi.org/10.1109/access.2022.3182372.

[27] F. Ariai, J. Mackenzie, and G. Demartini, "Natural language processing for the legal domain: A survey of tasks, datasets, models, and challenges," *ACM Computing Surveys*, vol. 58, no. 6, pp. 163:1–163:37, 2025. https://doi.org/10.1145/3777009.

[28] E. Aghaei, X. Niu, W. Shadid, and E. Al-Shaer, "SecureBERT: A domain-specific language model for cybersecurity." Security and Privacy in Communication Networks (SecureComm), 2022. https://doi.org/10.1007/978-3-031-25538-0_3.

[29] K. Ethayarajh, "How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 55–65, Association for Computational Linguistics, 2019. https://doi.org/10.18653/v1/D19-1006.

[30] S. Jamshidi, N. Shahabi, A. Nikanjam, K. W. Nafi, F. Khomh, and C. Fung, "The role of large language models in IoT security: A systematic review of advances, challenges, and opportunities," *Internet of Things*, vol. 34, p. 101735, 2025. https://doi.org/10.1016/j.iot.2025.101735.

[31] K. Sapkota, A. Aldea, M. Younas, D. A. Duce, and R. Banares-Alcantara, "Automating the semantic mapping between regulatory guidelines and organizational processes," *Service Oriented Computing and Applications*, vol. 10, no. 4, pp. 365–389, 2016. https://doi.org/10.1007/s11761-016-0197-2.

[32] F. Gilardi, M. Alizadeh, and M. Kubli, "ChatGPT outperforms crowd workers for text-annotation tasks," *Proceedings of the National Academy of Sciences*, vol. 120, no. 30, p. e2305016120, 2023. https://doi.org/10.1073/pnas.2305016120.

[33] C. Ziems, W. Held, O. Shaikh, J. Chen, Z. Zhang, and D. Yang, "Can large language models transform computational social science?," *Computational Linguistics*, vol. 50, no. 1, pp. 237–291, 2024. https://doi.org/10.1162/coli_a_00502.

[34] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*, pp. 46595–46623, 2023. https://doi.org/10.52202/075280-2020.

[35] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960. https://doi.org/10.1177/001316446002000104.

[36] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977. https://doi.org/10.2307/2529310.

[37] K. Song, X. Tan, T. Qin, J. Lu, and T.-Y. Liu, "MPNet: Masked and permuted pre-training for language understanding," in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, pp. 16857–16867, 2020.

[38] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, "C-Pack: Packed resources for general Chinese embeddings," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 641–649, 2024. https://doi.org/10.1145/3626772.3657878.

[39] M. Bayer, P. Kuehn, R. Shanehsaz, and C. Reuter, "CySecBERT: A domain-adapted language model for the cybersecurity domain," *ACM Transactions on Privacy and Security*, vol. 27, no. 2, pp. 1–20, 2024. https://doi.org/10.1145/3652594.

[40] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.

[41] A. Williams, N. Nangia, and S. Bowman, "A broad-coverage challenge corpus for sentence understanding through inference," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, pp. 1112–1122, Association for Computational Linguistics, 2018. https://doi.org/10.18653/v1/N18-1101.

Glossary and Symbols

Anisotropy	Property of an embedding space in which vectors occupy a narrow cone, so that cosine similarity between arbitrary items is high regardless of meaning.
BERT	Bidirectional Encoder Representations from Transformers; a pretrained contextual language model.
Bootstrap	Nonparametric resampling procedure used to estimate the sampling distribution of a metric, here for confidence intervals.
Cohen's kappa (κ)	Chance-corrected agreement measure between two raters on categorical labels.
Cosine similarity	Similarity of two vectors measured as the cosine of the angle between them; 1 means identical direction.
ETSI	European Telecommunications Standards Institute; publisher of the two standards studied.
F1 score	Harmonic mean of precision and recall.
Ground truth	The reference annotation against which automated predictions are evaluated; here drafted by one frontier LLM under a written codebook, verified in the later two rounds by a second from an unrelated developer, and reviewed by the authors, pair by pair for the pilot set.
IoT	Internet of Things; network-connected consumer and industrial devices.
LLM	Large Language Model; a generative transformer model used here for direct relationship classification.
Macro-F1	Unweighted mean of per-class F1 scores; treats rare and frequent classes equally.
Modality	The obligation level of a provision: shall (mandatory), should (recommended), or may (permitted).
NLP	Natural Language Processing.
Provision	A single numbered normative requirement in an ETSI standard.
SBERT	Sentence-BERT; a Siamese BERT architecture trained so that sentence-level cosine similarity reflects semantic relatedness.
TF–IDF	Term Frequency–Inverse Document Frequency; a lexical text weighting scheme.

A Relationship Label Definitions

The definitions below were used both for ground-truth annotation and in the prompt of the LLM classification method. a denotes a provision from ETSI EN 303 645, b a provision from ETSI EN 304 223.

Equivalence

a and b express the same security objective with comparable scope. Compliance with one gives strong evidence of compliance with the other. *Decision aid:* if an auditor would accept evidence for a as evidence for b without further argument, the pair is equivalent.

Subsumption (A broader)

a's objective fully contains b's: everything b requires is required by a, but a requires more. *Decision aid:* compliance with a implies compliance with b, not conversely.

Subsumption (B broader)

The mirror case: compliance with b implies compliance with a.

Overlap

The objectives intersect: part of what a requires is also required by b, but each provision also covers ground the other does not. Evidence can be partially reused; a gap analysis is still needed for the uncovered parts.

Complementarity

a and b address different aspects of a common security goal without overlapping scope — for example, one requires a technical mechanism and the other requires user-facing guidance about it. Implementing both is what produces the protection; neither substitutes for the other.

No relation

No meaningful connection between the objectives. Used both for obviously unrelated pairs and — more importantly for evaluation — for pairs that share vocabulary but not objective.

Annotation rules

Each pair receives exactly one label, a one-sentence justification, and a confidence score: 1 (uncertain), 2 (reasonably confident), 3 (certain). When several labels seem defensible, the annotator picks the most specific one that the provision *texts* (not assumed intent) support, and records the doubt in the justification.

B | Expert Validation Survey Instrument

This appendix contains the complete instrument and analysis plan for the proposed expert validation of the ground truth (Section 3.4.4). The study was designed but not executed within the degree project; it is specified here so it can be run without further design work. A distribution-ready copy is maintained in the project repository (`docs/expert_survey.md`).

B.1 Purpose

Measure (i) how consistently practising cybersecurity professionals assign the six relationship labels to provision pairs, and (ii) how well their majority judgement agrees with the project's ground truth. High agreement validates the ground truth as an evaluation reference; disagreement localises contestable annotations.

B.2 Participants and recruitment

Target group: professionals with at least two years of experience in cybersecurity engineering, compliance/GRC, or security assurance, familiar with at least one of the two standards or an equivalent framework (ISO/IEC 27001, IEC 62443). Target sample: 5–10 respondents recruited through professional networks and university industry contacts. Participation is anonymous and uncompensated.

B.3 Ethics and data handling

No personal data is collected beyond role category, years of experience, and self-rated familiarity. Responses are stored pseudonymously (R1, R2, . . .). Participants consent via a checkbox after reading a description of the study purpose and the intended publication of aggregated results. The design collects anonymous professional judgements and no sensitive data; supervisor confirmation is obtained before deployment.

B.4 Screening questions

S1. Current role: security engineer / compliance-GRC / auditor / researcher / other (single choice).

S2. Years of professional experience in cybersecurity (number; respondents under 2 years excluded from the main analysis).

S3. Familiarity with ETSI EN 303 645 (Likert 1–5).

S4. Familiarity with ETSI EN 304 223 or other AI-security guidance (Likert 1–5).

S5. Familiarity with compliance or standards mapping as an activity (Likert 1–5).

B.5 Main task

Each respondent rates the same 30 provision pairs drawn from the reference set, in per-respondent random order. The draw is stratified over the four labels with enough support to stratify on; the annotation found only three EQUIVALENCE pairs and ten SUBSUMPTION pairs in the whole corpus, so those two enter the instrument at whatever count is available rather than at an equal share. For each pair the full text of both provisions is shown together with the label definitions of Appendix A, and the respondent answers:

Q1. **Relationship** — exactly one of the six labels.

Q2. **Confidence** — 1 (guessing) to 5 (certain).

Q3. **Justification** — optional free text.

Estimated completion time is 35–45 minutes. One duplicated pair with reversed presentation order serves as an attention check; inconsistent answers flag the respondent for review rather than automatic exclusion.

B.6 Post-task questions

P1. Clarity of the six label definitions (Likert 1–5).

P2. Which pair of labels was hardest to distinguish? (choice among label pairs).

P3. Would a tool proposing candidate mappings with these labels help your compliance work? (Likert 1–5 plus free text).

B.7 Analysis plan

1. **Inter-expert agreement:** Fleiss' kappa over all respondents, on the six-label and the merged five-label scheme, interpreted with the Landis–Koch bands [36].
2. **Expert vs. ground truth:** per-item majority vote compared with the stored label; percent agreement and Cohen's kappa [35]. Items without a majority are reported as contested.
3. **Confidence:** mean confidence per label; point-biserial correlation between low confidence and disagreement.
4. **Ground-truth revision:** items where the expert majority disagrees with the stored label are re-examined; convincing corrections enter the ground truth through a documented change log, and the full evaluation is re-run to test whether the method ranking of Chapter 5 is stable under the revised reference.

B.8 Expected outcomes

Based on comparable annotation studies, moderate inter-expert agreement ($\kappa \approx 0.4$–0.6) is expected on the six-label scheme, higher after merging, with the overlap/complementarity boundary as the main source of disagreement. The ground truth is considered validated for its purpose if majority agreement reaches at least 70% on the merged scheme.

C | Reproduction Guide

All results in this thesis are regenerated by the following procedure. The project requires Python 3.13, the packages pinned in `requirements.txt` (scikit-learn 1.8, pandas 3.0, sentence-transformers 5.4, transformers 5.8, torch 2.11, google-genai 2.0, matplotlib 3.11), and Poppler for `pdftotext`.

Setup

```
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt
brew install poppler          # macOS; apt install poppler-utils on Debian
```

Pipeline

Run from the repository root, in order:

```
python3 -m src.extraction.provision_extractor   # ~1 s
python3 -m src.baseline.build_reference         # <1 s
python3 -m src.mapping.rule_based               # ~4 s
python3 -m src.mapping.run_embedding --all      # ~2 min, six models
python3 -m src.mapping.nli_mapping              # ~6 min
python3 -m src.evaluation.evaluate              # ~1 s
python3 -m src.evaluation.analysis              # ~1 min
python3 -m src.evaluation.ranking               # ~2 s
python3 -m src.evaluation.coverage              # ~2 s
python3 -m src.evaluation.supervised            # ~5 s
python3 -m src.report.generate_report           # ~1 s
python3 -m src.report.figures                   # ~15 s
```

The cross-encoder stage is the memory-sensitive one: it holds a batch of token activations on the accelerator, and on the 8 GB machine used here a batch larger than eight pushes the system into swap, where throughput falls by an order of magnitude. `NLI_BATCH_SIZE` and `NLI_DEVICE` override the defaults on hardware with more headroom.
Each embedding model can also be run on its own, `run_embedding <key>`; the keys are the ones listed in `src/methods.py`, which is the single place any method is defined. The earlier per-model entry points (`sbert_mapping`, `bert_mapping`, `securebert_mapping`) still work and call the same runner.

First runs of the transformer stages download the models from Hugging Face (about 2 GB in total; SecureBERT alone is 500 MB). The two Gemini stages,

```
python3 -m src.mapping.gemini_embedding_mapping
python3 -m src.mapping.gemini_mapping
```

require a `GEMINI_API_KEY` in `.env` and incur API cost; the repository ships their stored outputs, which the evaluation consumes directly. The run-to-run agreement figure of Table 5.7 needs a second pass of the classifier over byte-identical prompts, written to a separate file:

```
GEMINI_OUTPUT_SUFFIX=rep2 python3 -m src.mapping.gemini_mapping
```

Trailer ablation

The requirement-only condition of Section 5.3 re-scores the local methods on provision text stripped to its normative sentences:

```
python3 -m src.analysis.ablation_trailers --with-nli
```

Without `--with-nli` the cross-encoder is skipped and the run takes about two minutes. Results are written to `data/ablation/` and nothing under `data/extracted/` or `data/mappings/` is modified, so the ablation cannot disturb the figures reported for the full-text condition. The two hosted methods are archived as stored outputs and are absent from this comparison for that reason.

Reference set and reliability

```
python3 -m src.baseline.sample_reference    # screened round
python3 -m src.baseline.annotate_sample status
python3 -m src.baseline.rare_class_search status
python3 -m src.validation.agreement         # kappa + table
```

The annotation batches for all three rounds are stored under `data/annotation/`, so these steps rebuild the 200-pair reference set without re-annotating anything. Rebuilding a round from scratch instead requires the `emit` and `ingest` subcommands of `annotate_sample` (screened round) or `rare_class_search` (targeted round) once per batch.

Building the report

```
python3 -m src.report.check_thesis_numbers  # guard: tables vs. data
cd thesis
pdflatex main.tex && bibtex main && pdflatex main.tex && pdflatex main.tex
```

The guard re-reads every number printed in a results table directly from the stored evaluation files and exits non-zero on any mismatch. It is run before each submission build, and caught three transcription errors during writing.

Determinism

Sampling and bootstrap procedures use fixed seeds (7 and 42). A verification re-run during the project reproduced the stored evaluation summary exactly. Individual embedding scores can differ in the fourth decimal between runs (CPU thread scheduling inside the tensor library); the differences are far below the label thresholds and leave every reported metric unchanged. API-based results depend on hosted model versions and are preserved as stored outputs instead.

D | Predicted Provision Mapping

The table below is the mapping this project produces: every pair of provisions that the best-scoring method flags as related at its calibrated threshold, with the similarity score that produced the flag. It is included because it is the artefact a compliance practitioner would actually use, and because a comparative mapping study is expected to show its mapping rather than only summary statistics of it.

This is a predicted mapping, not a verified one. Every row is the output of an automated method whose precision on the annotated reference set was well below one; the measured value and its confidence interval are printed in Table 5.3 and in the caption of Figure 5.8. A substantial fraction of the rows below are therefore wrong. The table is a shortlist: the pairs an analyst would examine first, in the order the method ranks them. The measured statements in this thesis are those in Chapter 5, which carry intervals and are scored against annotated labels; nothing in this appendix is.

Rows are grouped by ETSI EN 303 645 provision and ordered within a group by descending score. The table is generated directly from the evaluation outputs by the `coverage` module, so it cannot drift from the data it summarises.

EN 303 645	EN 304 223	Score
5.0-1	5.1.3-1	0.751
5.0-1	5.2.3-2	0.749
5.0-1	5.2.1-1	0.742
5.0-1	5.1.2-1	0.741
5.0-1	5.1.3-1.1	0.734
5.0-1	5.1.2-7	0.727
5.0-1	5.2.3-2.1	0.725
5.0-1	5.1.3-2	0.725
5.0-1	5.1.2-3	0.723
5.0-1	5.2.4-1.1	0.722
5.0-1	5.2.2-4	0.722
5.1-1	5.4.1-1	0.733
5.1-1	5.2.1-2	0.732
5.1-1	5.2.1-4	0.729
5.1-1	5.2.5-1	0.727
5.1-1	5.2.3-1	0.720
5.1-1	5.1.3-3	0.720
5.1-1	5.1.2-7	0.719
5.1-1	5.2.1-1	0.719
5.1-1	5.2.2-4	0.718
5.1-1	5.2.1-4.1	0.717
5.1-1	5.2.2-1	0.716

EN 303 645	EN 304 223	Score
5.1-1	5.2.3-2.1	0.715
5.1-2	5.2.1-4	0.758
5.1-2	5.1.3-1	0.757
5.1-2	5.1.3-3	0.754
5.1-2	5.2.1-4.1	0.753
5.1-2	5.2.3-1	0.752
5.1-2	5.2.1-2	0.751
5.1-2	5.2.3-2.1	0.751
5.1-2	5.1.2-4	0.749
5.1-2	5.1.2-2	0.749
5.1-2	5.4.1-1	0.745
5.1-2	5.2.2-4	0.742
5.1-2	5.2.5-1	0.742
5.1-2	5.2.5-2.1	0.740
5.1-2	5.4.1-1.1	0.739
5.1-2	5.1.2-7	0.738
5.1-2	5.1.4-3	0.737
5.1-2	5.2.1-1	0.737
5.1-2	5.2.2-1	0.735
5.1-2	5.1.2-6	0.734
5.1-2	5.1.2-1	0.732
5.1-2	5.3.1-3	0.732
5.1-2	5.1.3-1.1	0.731
5.1-2	5.1.3-2	0.730
5.1-2	5.1.1-2.2	0.730
5.1-2	5.2.2-2	0.729
5.1-2	5.3.1-2.2	0.728
5.1-2	5.2.3-2	0.728
5.1-2	5.2.1-4.2	0.727
5.1-2	5.1.1-2	0.725
5.1-2	5.1.1-1	0.725
5.1-2	5.2.2-3	0.717
5.1-2	5.5.1-1	0.717
5.1-2	5.2.1-3	0.717
5.1-2	5.2.4-2	0.716
5.1-2	5.3.1-1	0.714
5.1-3	5.2.1-2	0.760
5.1-3	5.2.2-1	0.746
5.1-3	5.2.3-1	0.746
5.1-3	5.2.5-1	0.744
5.1-3	5.1.3-3	0.744
5.1-3	5.2.1-4	0.742
5.1-3	5.2.1-4.1	0.737
5.1-3	5.2.2-4	0.736
5.1-3	5.1.2-7	0.736
5.1-3	5.1.3-1	0.732
5.1-3	5.1.2-2	0.730
5.1-3	5.2.3-2.1	0.730

EN 303 645	EN 304 223	Score
5.1-3	5.4.1-1	0.729
5.1-3	5.2.5-2.1	0.728
5.1-3	5.1.2-1	0.727
5.1-3	5.1.2-4	0.724
5.1-3	5.3.1-3	0.724
5.1-3	5.2.3-2	0.723
5.1-3	5.2.4-1.2	0.722
5.1-3	5.2.1-1	0.719
5.1-3	5.2.2-6	0.718
5.1-3	5.1.1-2	0.718
5.1-3	5.2.2-2	0.715
5.1-3	5.2.1-4.2	0.715
5.1-3	5.4.2-1	0.715
5.1-3	5.1.2-6	0.715
5.1-4	5.4.1-1	0.746
5.1-4	5.3.1-2.2	0.743
5.1-4	5.2.2-4	0.740
5.1-4	5.2.1-4	0.728
5.1-4	5.2.5-1	0.724
5.1-4	5.2.1-2	0.721
5.1-4	5.1.3-3	0.716
5.1-4	5.3.1-3	0.715
5.1-4	5.2.1-1	0.715
5.1-4	5.1.3-1.1	0.714
5.1-4	5.5.1-1	0.714
5.1-4	5.2.3-1	0.714
5.1-5	5.1.2-2	0.745
5.1-5	5.2.1-4	0.744
5.1-5	5.2.1-2	0.743
5.1-5	5.2.2-1	0.742
5.1-5	5.2.2-2	0.741
5.1-5	5.1.3-1	0.740
5.1-5	5.2.5-1	0.736
5.1-5	5.1.3-3	0.726
5.1-5	5.2.2-4	0.726
5.1-5	5.2.3-1	0.723
5.1-5	5.1.2-1	0.723
5.1-5	5.2.1-4.1	0.718
5.1-5	5.2.1-1	0.717
5.1-5	5.1.1-2	0.716
5.10-1	5.4.2-3	0.776
5.10-1	5.4.2-4	0.746
5.10-1	5.4.2-2	0.745
5.10-1	5.4.1-2	0.714
5.11-1	5.5.1-2	0.733
5.11-1	5.3.1-1	0.730
5.11-2	5.5.1-2	0.761
5.11-2	5.3.1-1	0.721

EN 303 645	EN 304 223	Score
5.11-2	5.5.1-1	0.720
5.11-3	5.5.1-2	0.770
5.11-3	5.3.1-1	0.768
5.11-3	5.3.1-2.2	0.732
5.11-3	5.4.1-1	0.729
5.11-3	5.5.1-1	0.727
5.11-3	5.3.1-2	0.723
5.11-3	5.1.4-5	0.719
5.11-3	5.2.2-4	0.714
5.11-4	5.5.1-2	0.773
5.11-4	5.3.1-1	0.742
5.11-4	5.5.1-1	0.726
5.11-4	5.2.2-6	0.721
5.12-1	5.3.1-2.2	0.751
5.12-1	5.4.1-1	0.732
5.12-1	5.1.4-2	0.725
5.12-1	5.2.2-4	0.720
5.12-1	5.4.1-2	0.717
5.12-2	5.3.1-2.2	0.765
5.12-2	5.4.1-1	0.757
5.12-2	5.2.1-1	0.754
5.12-2	5.3.1-3	0.747
5.12-2	5.1.3-2	0.736
5.12-2	5.3.1-2	0.733
5.12-2	5.2.2-4	0.729
5.12-2	5.4.1-2	0.727
5.12-2	5.1.3-1.1	0.725
5.12-2	5.1.3-1	0.723
5.12-2	5.1.3-3	0.718
5.12-2	5.4.2-3	0.717
5.12-2	5.1.2-7	0.715
5.12-2	5.2.4-1.1	0.713
5.12-3	5.3.1-2.2	0.800
5.12-3	5.4.1-1	0.778
5.12-3	5.1.3-2	0.769
5.12-3	5.3.1-2	0.765
5.12-3	5.3.1-3	0.763
5.12-3	5.4.2-4	0.758
5.12-3	5.4.2-3	0.754
5.12-3	5.2.5-1	0.752
5.12-3	5.3.1-1	0.752
5.12-3	5.1.3-3	0.750
5.12-3	5.4.1-2	0.749
5.12-3	5.2.2-4	0.748
5.12-3	5.1.2-7	0.747
5.12-3	5.1.3-1	0.742
5.12-3	5.2.3-2.2	0.739
5.12-3	5.2.1-4.1	0.737

EN 303 645	EN 304 223	Score
5.12-3	5.4.1-1.1	0.736
5.12-3	5.1.4-2	0.735
5.12-3	5.2.1-1	0.734
5.12-3	5.2.4-1.1	0.731
5.12-3	5.1.3-4	0.730
5.12-3	5.1.3-1.1	0.729
5.12-3	5.2.3-1	0.729
5.12-3	5.1.1-2	0.729
5.12-3	5.3.1-2.1	0.729
5.12-3	5.2.5-2.1	0.728
5.12-3	5.2.5-4.1	0.728
5.12-3	5.4.2-2	0.728
5.12-3	5.1.4-1	0.728
5.12-3	5.1.2-1	0.727
5.12-3	5.1.1-1	0.724
5.12-3	5.1.2-4	0.723
5.12-3	5.2.1-2	0.721
5.12-3	5.2.3-2	0.721
5.12-3	5.1.4-3	0.719
5.12-3	5.2.4-1	0.718
5.12-3	5.1.4-5	0.718
5.12-3	5.2.3-4	0.718
5.12-3	5.2.1-4	0.717
5.12-3	5.2.5-3	0.716
5.12-3	5.1.4-4	0.715
5.13-1A	5.2.1-4.1	0.751
5.13-1A	5.2.1-4	0.736
5.13-1A	5.2.5-1	0.721
5.13-1A	5.1.4-4	0.717
5.13-1B	5.2.1-4.1	0.766
5.13-1B	5.2.5-1	0.746
5.13-1B	5.2.1-4	0.738
5.13-1B	5.1.2-2	0.730
5.13-1B	5.1.3-1	0.726
5.13-1B	5.2.1-1	0.726
5.13-1B	5.1.4-4	0.726
5.13-1B	5.2.2-4	0.715
5.2-1	5.2.2-4	0.871
5.2-1	5.4.1-1	0.778
5.2-1	5.1.3-1	0.767
5.2-1	5.2.5-1	0.767
5.2-1	5.4.1-1.1	0.766
5.2-1	5.2.1-1	0.763
5.2-1	5.2.3-1	0.763
5.2-1	5.1.3-2	0.759
5.2-1	5.3.1-3	0.759
5.2-1	5.1.3-3	0.758
5.2-1	5.3.1-2.2	0.757

EN 303 645	EN 304 223	Score
5.2-1	5.1.2-4	0.748
5.2-1	5.1.2-7	0.747
5.2-1	5.1.2-1	0.747
5.2-1	5.1.3-1.1	0.746
5.2-1	5.4.1-2	0.741
5.2-1	5.1.1-2	0.740
5.2-1	5.2.1-2	0.740
5.2-1	5.2.3-2	0.738
5.2-1	5.2.5-2.1	0.737
5.2-1	5.1.1-1	0.736
5.2-1	5.2.3-2.1	0.734
5.2-1	5.1.2-2	0.734
5.2-1	5.2.4-2	0.731
5.2-1	5.2.4-1.1	0.730
5.2-1	5.3.1-1	0.730
5.2-1	5.2.2-5	0.729
5.2-1	5.1.1-2.2	0.725
5.2-1	5.1.2-3	0.725
5.2-1	5.2.1-4.1	0.724
5.2-1	5.2.1-4	0.723
5.2-1	5.1.2-5.1	0.722
5.2-1	5.2.4-1	0.720
5.2-1	5.1.4-3	0.718
5.2-1	5.4.2-1	0.717
5.2-1	5.1.3-4	0.716
5.2-1	5.2.2-2	0.716
5.2-1	5.4.2-3	0.715
5.2-1	5.2.3-2.2	0.713
5.2-2	5.2.2-4	0.781
5.2-2	5.4.1-1	0.728
5.2-2	5.2.3-2	0.717
5.2-3	5.4.1-1	0.787
5.2-3	5.2.2-4	0.786
5.2-3	5.1.3-1	0.774
5.2-3	5.3.1-3	0.767
5.2-3	5.4.2-4	0.766
5.2-3	5.4.2-3	0.764
5.2-3	5.1.1-2	0.760
5.2-3	5.4.1-2	0.759
5.2-3	5.4.1-1.1	0.756
5.2-3	5.3.1-2.2	0.755
5.2-3	5.1.3-4	0.754
5.2-3	5.2.5-1	0.751
5.2-3	5.2.1-1	0.744
5.2-3	5.1.3-2	0.742
5.2-3	5.2.3-1	0.742
5.2-3	5.1.3-3	0.741
5.2-3	5.1.1-1	0.738

EN 303 645	EN 304 223	Score
5.2-3	5.1.3-1.1	0.738
5.2-3	5.2.1-4.1	0.734
5.2-3	5.1.1-2.2	0.734
5.2-3	5.1.2-7	0.732
5.2-3	5.1.2-2	0.732
5.2-3	5.2.1-2	0.730
5.2-3	5.1.2-4	0.728
5.2-3	5.2.5-2.1	0.724
5.2-3	5.4.2-2	0.723
5.2-3	5.1.2-1	0.718
5.2-3	5.4.2-1	0.716
5.2-3	5.1.4-1	0.714
5.2-3	5.2.3-2	0.713
5.3-1	5.4.1-1	0.790
5.3-1	5.3.1-2.2	0.776
5.3-1	5.4.1-2	0.760
5.3-1	5.2.3-1	0.758
5.3-1	5.4.1-1.1	0.753
5.3-1	5.2.2-4	0.748
5.3-1	5.2.5-1	0.722
5.3-1	5.1.3-1	0.721
5.3-1	5.2.1-1	0.720
5.3-1	5.1.2-4	0.719
5.3-1	5.1.1-1	0.715
5.3-1	5.2.1-2	0.715
5.3-1	5.1.3-1.1	0.714
5.3-10	5.4.1-1	0.786
5.3-10	5.2.3-1	0.774
5.3-10	5.3.1-2.2	0.762
5.3-10	5.1.3-3	0.758
5.3-10	5.2.5-1	0.755
5.3-10	5.1.2-4	0.752
5.3-10	5.4.1-1.1	0.751
5.3-10	5.4.1-2	0.749
5.3-10	5.2.4-1.2	0.745
5.3-10	5.2.1-4.1	0.743
5.3-10	5.2.2-4	0.743
5.3-10	5.1.3-1	0.743
5.3-10	5.2.1-2	0.738
5.3-10	5.1.2-2	0.737
5.3-10	5.1.2-7	0.736
5.3-10	5.2.3-2.1	0.735
5.3-10	5.1.1-1	0.734
5.3-10	5.2.1-1	0.731
5.3-10	5.1.3-2	0.728
5.3-10	5.1.1-2	0.726
5.3-10	5.2.1-4	0.726
5.3-10	5.1.4-3	0.725

EN 303 645	EN 304 223	Score
5.3-10	5.1.3-1.1	0.724
5.3-10	5.2.3-2	0.722
5.3-10	5.3.1-3	0.722
5.3-10	5.2.1-4.2	0.721
5.3-10	5.5.1-1	0.721
5.3-10	5.2.5-2.1	0.719
5.3-11	5.3.1-2.2	0.814
5.3-11	5.4.1-1	0.799
5.3-11	5.4.1-2	0.749
5.3-11	5.2.2-4	0.747
5.3-11	5.1.3-2	0.740
5.3-11	5.1.1-2.1	0.737
5.3-11	5.4.1-1.1	0.735
5.3-11	5.2.3-4	0.717
5.3-12	5.3.1-2.2	0.759
5.3-12	5.2.3-4	0.754
5.3-12	5.4.1-1	0.737
5.3-13	5.3.1-2.2	0.814
5.3-13	5.4.1-1	0.802
5.3-13	5.2.2-4	0.788
5.3-13	5.3.1-1	0.772
5.3-13	5.2.3-4	0.767
5.3-13	5.2.3-2.2	0.766
5.3-13	5.3.1-2	0.764
5.3-13	5.3.1-3	0.751
5.3-13	5.4.1-1.1	0.750
5.3-13	5.2.3-1	0.745
5.3-13	5.2.5-1	0.735
5.3-13	5.1.3-2	0.734
5.3-13	5.2.3-2	0.733
5.3-13	5.1.2-7	0.732
5.3-13	5.4.1-2	0.729
5.3-13	5.1.3-1	0.729
5.3-13	5.2.1-1	0.727
5.3-13	5.1.2-4	0.726
5.3-13	5.1.2-2	0.724
5.3-13	5.4.1-3	0.722
5.3-13	5.1.4-5	0.721
5.3-13	5.2.4-1	0.720
5.3-13	5.1.2-3	0.720
5.3-13	5.2.2-5	0.715
5.3-13	5.2.2-6	0.714
5.3-13	5.1.4-2	0.713
5.3-14	5.3.1-2.2	0.813
5.3-14	5.2.2-4	0.790
5.3-14	5.4.1-1	0.790
5.3-14	5.3.1-1	0.771
5.3-14	5.4.1-1.1	0.766

EN 303 645	EN 304 223	Score
5.3-14	5.3.1-3	0.766
5.3-14	5.3.1-2	0.765
5.3-14	5.2.3-2.2	0.760
5.3-14	5.2.3-4	0.754
5.3-14	5.1.3-2	0.745
5.3-14	5.2.3-1	0.743
5.3-14	5.2.3-2	0.740
5.3-14	5.2.1-1	0.736
5.3-14	5.4.1-2	0.733
5.3-14	5.1.2-4	0.730
5.3-14	5.1.2-7	0.730
5.3-14	5.2.2-6	0.727
5.3-14	5.1.4-5	0.726
5.3-14	5.2.4-1	0.725
5.3-14	5.2.3-2.1	0.724
5.3-14	5.1.2-2	0.723
5.3-14	5.1.3-3	0.723
5.3-14	5.1.4-2	0.723
5.3-14	5.4.1-3	0.720
5.3-14	5.2.5-1	0.720
5.3-14	5.1.3-1	0.720
5.3-14	5.2.1-4.1	0.718
5.3-14	5.2.4-1.1	0.717
5.3-14	5.1.2-3	0.716
5.3-14	5.2.5-3	0.714
5.3-14	5.1.1-1	0.714
5.3-14	5.1.3-4	0.713
5.3-15B	5.4.1-1.1	0.749
5.3-15B	5.4.1-1	0.749
5.3-15B	5.3.1-2.2	0.732
5.3-15B	5.4.1-2	0.732
5.3-15B	5.2.2-4	0.723
5.3-16	5.2.5-1	0.733
5.3-16	5.2.3-4	0.731
5.3-16	5.3.1-1	0.730
5.3-16	5.4.1-3	0.725
5.3-16	5.4.1-1	0.723
5.3-16	5.2.3-2.1	0.721
5.3-16	5.3.1-2.2	0.719
5.3-16	5.2.1-1	0.719
5.3-16	5.1.4-2	0.719
5.3-16	5.2.5-4.1	0.719
5.3-16	5.2.2-4	0.717
5.3-16	5.1.2-7	0.715
5.3-16	5.2.4-1.2	0.715
5.3-16	5.2.3-2	0.714
5.3-16	5.2.3-3	0.714
5.3-16	5.2.3-2.2	0.713

EN 303 645	EN 304 223	Score
5.3-2	5.4.1-1	0.781
5.3-2	5.4.1-1.1	0.770
5.3-2	5.3.1-2.2	0.765
5.3-2	5.2.3-1	0.762
5.3-2	5.2.2-4	0.759
5.3-2	5.2.5-1	0.740
5.3-2	5.4.1-2	0.739
5.3-2	5.2.3-2.1	0.738
5.3-2	5.2.3-2	0.734
5.3-2	5.1.2-4	0.732
5.3-2	5.1.3-1.1	0.730
5.3-2	5.2.1-2	0.730
5.3-2	5.1.2-1	0.729
5.3-2	5.1.3-1	0.729
5.3-2	5.1.1-2	0.729
5.3-2	5.2.1-1	0.727
5.3-2	5.1.1-1	0.726
5.3-2	5.1.3-3	0.724
5.3-2	5.1.3-2	0.720
5.3-2	5.2.1-4	0.718
5.3-2	5.2.1-4.1	0.718
5.3-2	5.1.2-7	0.717
5.3-2	5.1.1-2.2	0.714
5.3-3	5.3.1-2.2	0.775
5.3-3	5.4.1-1	0.774
5.3-3	5.2.2-4	0.729
5.3-3	5.4.1-2	0.717
5.3-3	5.1.4-2	0.717
5.3-3	5.3.1-1	0.716
5.3-3	5.2.3-2.2	0.715
5.3-4A	5.4.1-1	0.750
5.3-4A	5.2.1-3.1	0.739
5.3-4A	5.3.1-2.2	0.729
5.3-4A	5.2.2-4	0.725
5.3-4A	5.4.1-2	0.722
5.3-4A	5.2.3-1	0.718
5.3-4B	5.4.1-1	0.788
5.3-4B	5.3.1-2.2	0.774
5.3-4B	5.4.1-1.1	0.768
5.3-4B	5.4.1-2	0.765
5.3-4B	5.3.1-3	0.753
5.3-4B	5.2.2-4	0.747
5.3-4B	5.2.1-3.1	0.740
5.3-4B	5.1.3-1	0.734
5.3-4B	5.1.3-1.1	0.731
5.3-4B	5.2.3-1	0.730
5.3-4B	5.1.2-2	0.727
5.3-4B	5.2.2-5	0.726

EN 303 645	EN 304 223	Score
5.3-4B	5.1.3-2	0.726
5.3-4B	5.1.2-4	0.724
5.3-4B	5.2.1-4.1	0.722
5.3-4B	5.1.1-1	0.720
5.3-4B	5.2.1-4	0.718
5.3-4B	5.2.5-1	0.717
5.3-4B	5.2.1-3	0.716
5.3-4B	5.2.1-1	0.716
5.3-4B	5.4.2-3	0.715
5.3-4B	5.1.4-3	0.715
5.3-4B	5.2.3-2.1	0.715
5.3-4B	5.1.4-1	0.714
5.3-4B	5.1.1-2	0.713
5.3-5	5.4.1-1	0.761
5.3-5	5.4.1-2	0.754
5.3-5	5.3.1-2.2	0.748
5.3-5	5.2.2-4	0.724
5.3-5	5.4.1-1.1	0.724
5.3-6A	5.4.1-1	0.768
5.3-6A	5.3.1-2.2	0.762
5.3-6A	5.4.1-2	0.729
5.3-6A	5.4.1-1.1	0.721
5.3-6A	5.2.2-4	0.717
5.3-6B	5.3.1-2.2	0.769
5.3-6B	5.4.1-1	0.760
5.3-6B	5.3.1-1	0.721
5.3-6B	5.2.3-4	0.719
5.3-6B	5.4.1-2	0.715
5.3-6B	5.1.4-1	0.714
5.3-7	5.4.1-1	0.777
5.3-7	5.2.3-1	0.760
5.3-7	5.3.1-2.2	0.758
5.3-7	5.4.1-1.1	0.740
5.3-7	5.2.5-1	0.738
5.3-7	5.2.2-4	0.737
5.3-7	5.4.1-2	0.736
5.3-7	5.1.3-1	0.731
5.3-7	5.2.4-1.2	0.727
5.3-7	5.2.1-2	0.724
5.3-7	5.1.3-1.1	0.721
5.3-7	5.1.2-2	0.715
5.3-7	5.3.1-3	0.715
5.3-7	5.1.2-7	0.713
5.3-8	5.4.1-1	0.802
5.3-8	5.3.1-2.2	0.790
5.3-8	5.2.2-4	0.754
5.3-8	5.4.1-1.1	0.744
5.3-8	5.4.1-2	0.744

EN 303 645	EN 304 223	Score
5.3-8	5.1.3-2	0.738
5.3-8	5.1.1-2.1	0.729
5.3-8	5.3.1-3	0.720
5.3-8	5.2.5-1	0.715
5.3-8	5.1.3-1	0.715
5.3-9	5.4.1-2	0.736
5.3-9	5.4.1-1	0.731
5.3-9	5.2.4-1.2	0.731
5.3-9	5.2.3-1	0.718
5.4-1	5.2.1-4	0.772
5.4-1	5.2.1-2	0.746
5.4-1	5.2.3-1	0.736
5.4-1	5.2.1-4.2	0.733
5.4-1	5.2.1-1	0.732
5.4-1	5.2.2-4	0.729
5.4-1	5.4.1-1	0.724
5.4-1	5.1.3-1	0.724
5.4-1	5.2.5-1	0.723
5.4-1	5.2.2-6	0.718
5.4-1	5.2.2-1	0.718
5.4-2	5.1.2-2	0.758
5.4-2	5.2.1-2	0.758
5.4-2	5.2.3-1	0.756
5.4-2	5.2.1-4	0.750
5.4-2	5.1.3-1	0.745
5.4-2	5.2.2-4	0.740
5.4-2	5.1.1-2.2	0.740
5.4-2	5.4.1-1	0.740
5.4-2	5.2.1-4.1	0.739
5.4-2	5.2.1-4.2	0.737
5.4-2	5.2.5-1	0.737
5.4-2	5.2.1-1	0.734
5.4-2	5.2.2-1	0.732
5.4-2	5.1.3-3	0.730
5.4-2	5.4.1-1.1	0.726
5.4-2	5.1.1-2	0.724
5.4-2	5.1.2-1	0.723
5.4-2	5.2.2-3	0.723
5.4-2	5.4.2-3	0.722
5.4-2	5.2.5-2.1	0.722
5.4-2	5.3.1-3	0.721
5.4-2	5.2.3-2.1	0.721
5.4-2	5.1.2-4	0.720
5.4-2	5.5.1-1	0.720
5.4-2	5.1.1-1	0.719
5.4-2	5.1.2-7	0.719
5.4-2	5.2.2-6	0.718
5.4-2	5.1.4-3	0.715

EN 303 645	EN 304 223	Score
5.4-2	5.1.3-2	0.714
5.4-3	5.2.3-1	0.751
5.4-3	5.2.5-1	0.751
5.4-3	5.2.1-2	0.748
5.4-3	5.1.1-2.2	0.739
5.4-3	5.2.2-4	0.737
5.4-3	5.1.3-1	0.728
5.4-3	5.4.1-1	0.723
5.4-3	5.1.2-7	0.722
5.4-3	5.2.1-4	0.720
5.4-3	5.2.1-4.1	0.714
5.4-3	5.1.2-4	0.714
5.4-3	5.1.4-4	0.713
5.4-3	5.2.2-3	0.713
5.4-4	5.2.3-1	0.790
5.4-4	5.4.1-1	0.783
5.4-4	5.1.3-1	0.773
5.4-4	5.2.1-2	0.771
5.4-4	5.4.1-1.1	0.764
5.4-4	5.3.1-2.2	0.764
5.4-4	5.1.2-2	0.762
5.4-4	5.2.5-1	0.762
5.4-4	5.1.2-4	0.757
5.4-4	5.2.1-4.1	0.757
5.4-4	5.2.2-4	0.756
5.4-4	5.2.1-4	0.753
5.4-4	5.2.3-2.1	0.751
5.4-4	5.1.3-1.1	0.750
5.4-4	5.1.1-2.2	0.748
5.4-4	5.1.1-1	0.748
5.4-4	5.1.3-3	0.748
5.4-4	5.4.1-2	0.748
5.4-4	5.2.1-1	0.746
5.4-4	5.2.5-2.1	0.744
5.4-4	5.2.3-2	0.744
5.4-4	5.1.1-2	0.743
5.4-4	5.1.3-2	0.743
5.4-4	5.1.4-3	0.743
5.4-4	5.3.1-3	0.739
5.4-4	5.1.2-1	0.735
5.4-4	5.1.2-7	0.733
5.4-4	5.2.1-4.2	0.731
5.4-4	5.1.2-3	0.727
5.4-4	5.2.1-3	0.723
5.4-4	5.2.2-1	0.722
5.4-4	5.2.2-6	0.722
5.4-4	5.4.2-3	0.722
5.4-4	5.2.2-2	0.721

EN 303 645	EN 304 223	Score
5.4-4	5.2.4-1.2	0.720
5.4-4	5.2.4-2	0.720
5.4-4	5.2.2-3	0.719
5.4-4	5.5.1-1	0.718
5.4-4	5.2.2-5	0.718
5.4-4	5.4.2-4	0.717
5.4-4	5.1.2-6	0.717
5.4-4	5.1.3-4	0.716
5.4-4	5.3.1-1	0.715
5.4-4	5.1.3-1.3	0.714
5.4-4	5.4.2-1	0.713
5.5-1	5.1.2-7	0.744
5.5-1	5.2.3-1	0.742
5.5-1	5.2.5-1	0.742
5.5-1	5.4.1-1	0.735
5.5-1	5.2.2-6	0.734
5.5-1	5.2.2-4	0.733
5.5-1	5.1.3-3	0.732
5.5-1	5.3.1-3	0.730
5.5-1	5.1.3-1	0.729
5.5-1	5.3.1-2.2	0.727
5.5-1	5.2.1-2	0.727
5.5-1	5.2.3-2.1	0.723
5.5-1	5.2.3-2	0.720
5.5-1	5.2.1-4.2	0.719
5.5-1	5.2.1-4.1	0.718
5.5-1	5.2.5-2.1	0.718
5.5-1	5.1.2-4	0.717
5.5-1	5.2.4-1.2	0.716
5.5-1	5.2.2-1	0.715
5.5-1	5.3.1-1	0.715
5.5-1	5.4.2-3	0.715
5.5-1	5.2.2-2	0.714
5.5-1	5.2.1-4	0.714
5.5-1	5.1.3-2	0.713
5.5-2	5.2.5-1	0.768
5.5-2	5.1.2-4	0.763
5.5-2	5.2.3-1	0.762
5.5-2	5.1.2-7	0.760
5.5-2	5.2.2-4	0.758
5.5-2	5.1.3-3	0.755
5.5-2	5.2.5-2.1	0.750
5.5-2	5.1.3-1	0.745
5.5-2	5.2.3-2	0.745
5.5-2	5.2.1-2	0.744
5.5-2	5.4.1-1	0.739
5.5-2	5.2.3-2.1	0.739
5.5-2	5.4.1-2	0.737

EN 303 645	EN 304 223	Score
5.5-2	5.2.4-1.2	0.732
5.5-2	5.2.1-1	0.731
5.5-2	5.4.2-3	0.731
5.5-2	5.1.4-4	0.728
5.5-2	5.2.2-1	0.724
5.5-2	5.3.1-3	0.724
5.5-2	5.2.1-4.1	0.724
5.5-2	5.2.2-6	0.722
5.5-2	5.1.1-2.2	0.721
5.5-2	5.2.5-3	0.721
5.5-2	5.1.2-2	0.720
5.5-2	5.4.1-1.1	0.720
5.5-2	5.1.2-1	0.719
5.5-2	5.1.1-1	0.719
5.5-2	5.2.3-3	0.714
5.5-2	5.1.3-4	0.714
5.5-2	5.2.4-1.1	0.713
5.5-2	5.4.2-4	0.713
5.5-5	5.2.1-4	0.756
5.5-5	5.2.1-2	0.752
5.5-5	5.2.5-1	0.752
5.5-5	5.2.2-1	0.752
5.5-5	5.1.3-1.1	0.752
5.5-5	5.1.3-1	0.740
5.5-5	5.3.1-2.2	0.738
5.5-5	5.2.2-4	0.736
5.5-5	5.4.1-1	0.735
5.5-5	5.1.3-3	0.735
5.5-5	5.2.1-1	0.733
5.5-5	5.2.3-1	0.730
5.5-5	5.3.1-3	0.730
5.5-5	5.1.1-1	0.729
5.5-5	5.2.1-4.2	0.728
5.5-5	5.2.1-4.1	0.726
5.5-5	5.1.1-2	0.725
5.5-5	5.1.2-2	0.725
5.5-5	5.1.2-1	0.725
5.5-5	5.2.3-2	0.725
5.5-5	5.2.2-2	0.721
5.5-5	5.1.2-6	0.719
5.5-5	5.2.5-2.1	0.717
5.5-5	5.1.2-7	0.717
5.5-5	5.4.1-1.1	0.716
5.5-5	5.2.2-3	0.715
5.5-5	5.2.3-2.1	0.715
5.5-5	5.4.1-2	0.715
5.5-5	5.5.1-1	0.714
5.5-5	5.1.3-2	0.714

EN 303 645	EN 304 223	Score
5.5-5	5.1.1-2.2	0.714
5.5-6	5.2.1-4	0.743
5.5-6	5.1.3-3	0.736
5.5-6	5.1.4-4	0.728
5.5-6	5.2.2-6	0.725
5.5-6	5.2.1-2	0.725
5.5-6	5.2.1-4.2	0.723
5.5-6	5.5.1-1	0.721
5.5-6	5.2.5-1	0.720
5.5-6	5.1.3-1	0.718
5.5-7	5.2.1-4	0.754
5.5-7	5.2.1-4.2	0.752
5.5-7	5.1.3-1	0.728
5.5-7	5.2.2-4	0.725
5.5-7	5.2.5-1	0.722
5.5-7	5.1.3-3	0.719
5.5-7	5.2.2-2	0.717
5.5-7	5.1.3-2	0.716
5.5-7	5.2.1-2	0.716
5.5-7	5.2.2-1	0.715
5.5-7	5.1.2-7	0.714
5.5-7	5.3.1-2.2	0.713
5.5-7	5.1.1-2	0.713
5.5-8	5.2.3-1	0.789
5.5-8	5.1.3-1	0.777
5.5-8	5.2.1-2	0.775
5.5-8	5.1.3-1.1	0.773
5.5-8	5.2.2-4	0.763
5.5-8	5.4.1-1	0.762
5.5-8	5.2.1-1	0.759
5.5-8	5.2.5-1	0.757
5.5-8	5.2.1-4	0.754
5.5-8	5.4.1-1.1	0.745
5.5-8	5.1.3-3	0.744
5.5-8	5.3.1-3	0.741
5.5-8	5.1.2-7	0.741
5.5-8	5.5.1-1	0.737
5.5-8	5.2.2-1	0.736
5.5-8	5.1.3-2	0.735
5.5-8	5.1.2-1	0.734
5.5-8	5.1.1-2.2	0.734
5.5-8	5.1.1-1	0.734
5.5-8	5.3.1-2.2	0.733
5.5-8	5.1.1-2	0.733
5.5-8	5.2.1-4.2	0.732
5.5-8	5.1.2-3	0.732
5.5-8	5.4.2-3	0.732
5.5-8	5.1.2-4	0.731

EN 303 645	EN 304 223	Score
5.5-8	5.2.2-5	0.729
5.5-8	5.2.3-2	0.728
5.5-8	5.2.3-2.1	0.728
5.5-8	5.2.2-6	0.727
5.5-8	5.2.4-1.1	0.727
5.5-8	5.2.1-4.1	0.725
5.5-8	5.1.2-2	0.725
5.5-8	5.2.5-2.1	0.725
5.5-8	5.4.1-2	0.722
5.5-8	5.4.2-4	0.721
5.5-8	5.4.2-1	0.721
5.5-8	5.2.2-3	0.719
5.5-8	5.2.4-1	0.717
5.5-8	5.1.3-1.3	0.714
5.6-1	5.1.3-1.2	0.747
5.6-1	5.2.2-4	0.735
5.6-1	5.2.5-1	0.726
5.6-1	5.2.1-4	0.716
5.6-1	5.4.1-1	0.716
5.6-1	5.1.2-7	0.714
5.6-2	5.2.1-4	0.754
5.6-2	5.2.2-4	0.750
5.6-2	5.2.1-1	0.734
5.6-2	5.2.5-1	0.734
5.6-2	5.1.3-1.2	0.732
5.6-2	5.1.3-1	0.732
5.6-2	5.2.1-4.2	0.730
5.6-2	5.3.1-2.2	0.730
5.6-2	5.2.5-4	0.729
5.6-2	5.1.2-7	0.724
5.6-2	5.1.3-2	0.723
5.6-2	5.3.1-1	0.723
5.6-2	5.2.2-1	0.723
5.6-2	5.2.3-1	0.722
5.6-2	5.4.2-3	0.721
5.6-2	5.2.1-4.1	0.721
5.6-2	5.1.2-1	0.720
5.6-2	5.2.5-4.1	0.719
5.6-2	5.1.3-3	0.718
5.6-2	5.5.1-1	0.717
5.6-2	5.2.1-2	0.717
5.6-2	5.2.2-2	0.716
5.6-2	5.1.2-6	0.714
5.6-2	5.4.1-1	0.713
5.6-3	5.2.2-4	0.729
5.6-3	5.1.3-1.2	0.725
5.6-3	5.2.5-1	0.725
5.6-4A	5.2.5-1	0.760

EN 303 645	EN 304 223	Score
5.6-4A	5.2.1-4	0.754
5.6-4A	5.2.2-4	0.748
5.6-4A	5.2.1-2	0.743
5.6-4A	5.2.3-1	0.741
5.6-4A	5.2.2-1	0.740
5.6-4A	5.1.3-1	0.736
5.6-4A	5.4.1-1	0.734
5.6-4A	5.1.2-7	0.729
5.6-4A	5.1.2-2	0.727
5.6-4A	5.2.3-2	0.727
5.6-4A	5.2.2-3	0.726
5.6-4A	5.2.1-4.1	0.726
5.6-4A	5.1.2-4	0.726
5.6-4A	5.4.1-1.1	0.724
5.6-4A	5.2.2-2	0.724
5.6-4A	5.1.2-3	0.723
5.6-4A	5.1.1-2.2	0.722
5.6-4A	5.4.2-3	0.722
5.6-4A	5.1.3-1.1	0.720
5.6-4A	5.2.3-2.1	0.719
5.6-4A	5.1.3-1.2	0.719
5.6-4A	5.2.1-4.2	0.718
5.6-4A	5.2.1-1	0.718
5.6-4A	5.3.1-3	0.718
5.6-4A	5.3.1-2.2	0.718
5.6-4A	5.1.2-6	0.717
5.6-4A	5.1.4-3	0.716
5.6-4A	5.1.2-1	0.716
5.6-4A	5.1.3-2	0.715
5.6-4A	5.1.3-3	0.713
5.6-5	5.1.3-1.2	0.755
5.6-5	5.2.3-1	0.729
5.6-5	5.2.2-4	0.722
5.6-5	5.2.1-1	0.720
5.6-5	5.4.1-1	0.718
5.6-5	5.1.2-7	0.716
5.6-5	5.1.3-1	0.716
5.6-5	5.4.2-3	0.715
5.6-7	5.1.2-6	0.768
5.6-7	5.1.3-1.2	0.737
5.6-7	5.2.2-1	0.728
5.6-7	5.2.2-3	0.725
5.6-7	5.2.1-2	0.722
5.6-7	5.1.3-1	0.715
5.6-8	5.2.2-1	0.769
5.6-8	5.2.1-2	0.746
5.6-8	5.2.1-4	0.743
5.6-8	5.2.2-3	0.733

EN 303 645	EN 304 223	Score
5.6-8	5.1.3-1	0.729
5.6-8	5.2.3-1	0.726
5.6-8	5.2.5-1	0.724
5.6-8	5.1.2-2	0.722
5.6-8	5.2.2-2	0.722
5.6-8	5.1.1-2.2	0.721
5.6-8	5.2.2-4	0.720
5.6-8	5.1.3-1.1	0.717
5.6-8	5.2.1-1	0.717
5.6-8	5.1.2-1	0.717
5.6-8	5.2.1-4.2	0.716
5.6-8	5.2.3-2.1	0.715
5.6-8	5.4.1-1.1	0.714
5.6-8	5.4.2-3	0.713
5.6-8	5.4.1-1	0.713
5.6-9	5.2.3-1	0.821
5.6-9	5.2.1-2	0.802
5.6-9	5.2.5-1	0.784
5.6-9	5.1.3-1	0.776
5.6-9	5.2.2-4	0.772
5.6-9	5.1.1-2.2	0.770
5.6-9	5.4.1-1	0.770
5.6-9	5.1.2-7	0.769
5.6-9	5.4.1-2	0.767
5.6-9	5.2.4-1	0.760
5.6-9	5.1.2-4	0.753
5.6-9	5.2.1-4.1	0.752
5.6-9	5.1.3-1.1	0.752
5.6-9	5.3.1-3	0.751
5.6-9	5.1.2-3	0.750
5.6-9	5.4.1-1.1	0.750
5.6-9	5.2.1-1	0.749
5.6-9	5.2.3-2	0.744
5.6-9	5.2.3-2.1	0.743
5.6-9	5.2.2-6	0.741
5.6-9	5.2.4-3	0.741
5.6-9	5.4.2-3	0.740
5.6-9	5.2.4-1.1	0.738
5.6-9	5.2.1-4.2	0.734
5.6-9	5.2.2-3	0.733
5.6-9	5.2.2-1	0.731
5.6-9	5.1.2-2	0.730
5.6-9	5.2.5-3	0.729
5.6-9	5.1.4-2	0.728
5.6-9	5.1.2-1	0.727
5.6-9	5.2.2-2	0.726
5.6-9	5.2.4-2	0.726
5.6-9	5.1.1-2	0.725

EN 303 645	EN 304 223	Score
5.6-9	5.1.3-4	0.725
5.6-9	5.2.5-4.1	0.722
5.6-9	5.1.4-4	0.722
5.6-9	5.4.2-4	0.721
5.6-9	5.2.5-2.1	0.720
5.6-9	5.2.4-1.2	0.719
5.6-9	5.1.3-1.2	0.719
5.6-9	5.2.1-4	0.718
5.6-9	5.1.3-2	0.718
5.6-9	5.1.3-3	0.715
5.6-9	5.1.1-1	0.714
5.6-9	5.3.1-2.2	0.714
5.7-1	5.2.3-1	0.746
5.7-1	5.1.3-3	0.730
5.7-1	5.1.2-2	0.728
5.7-1	5.2.1-2	0.724
5.7-1	5.2.5-1	0.716
5.7-1	5.1.2-4	0.715
5.7-2	5.3.1-3	0.757
5.7-2	5.4.1-1	0.751
5.7-2	5.4.1-2	0.745
5.7-2	5.2.1-3.1	0.741
5.7-2	5.4.2-3	0.739
5.7-2	5.1.3-2	0.738
5.7-2	5.3.1-2.2	0.738
5.7-2	5.2.1-4	0.737
5.7-2	5.4.1-1.1	0.734
5.7-2	5.1.2-2	0.731
5.7-2	5.2.1-4.1	0.731
5.7-2	5.1.3-1	0.729
5.7-2	5.4.2-4	0.728
5.7-2	5.1.3-3	0.725
5.7-2	5.2.3-2.1	0.725
5.7-2	5.2.3-1	0.723
5.7-2	5.2.3-2	0.723
5.7-2	5.1.3-4	0.722
5.7-2	5.2.1-3	0.721
5.7-2	5.1.2-4	0.721
5.7-2	5.2.1-2	0.720
5.7-2	5.2.2-4	0.718
5.7-2	5.1.3-1.1	0.717
5.7-2	5.2.5-1	0.716
5.7-2	5.1.1-2	0.715
5.7-2	5.2.2-5	0.715
5.7-2	5.1.1-1	0.715
5.8-1	5.2.1-4.2	0.778
5.8-1	5.2.1-4	0.769
5.8-1	5.2.2-6	0.755

EN 303 645	EN 304 223	Score
5.8-1	5.3.1-1	0.742
5.8-1	5.2.2-4	0.739
5.8-1	5.1.3-3	0.735
5.8-1	5.2.1-2	0.734
5.8-1	5.2.5-1	0.734
5.8-1	5.1.3-1	0.732
5.8-1	5.2.3-1	0.732
5.8-1	5.1.2-7	0.729
5.8-1	5.3.1-3	0.728
5.8-1	5.4.1-1	0.728
5.8-1	5.2.1-1	0.728
5.8-1	5.5.1-1	0.725
5.8-1	5.2.2-1	0.722
5.8-1	5.2.1-4.1	0.720
5.8-1	5.1.4-4	0.718
5.8-1	5.3.1-2.2	0.717
5.8-1	5.4.2-3	0.715
5.8-1	5.1.1-2	0.714
5.8-2	5.2.1-4.2	0.782
5.8-2	5.2.1-4	0.782
5.8-2	5.2.2-6	0.756
5.8-2	5.3.1-1	0.745
5.8-2	5.2.2-4	0.745
5.8-2	5.2.5-1	0.745
5.8-2	5.2.3-1	0.741
5.8-2	5.2.1-1	0.740
5.8-2	5.2.1-2	0.740
5.8-2	5.1.3-1	0.739
5.8-2	5.4.1-1	0.739
5.8-2	5.1.3-3	0.736
5.8-2	5.1.2-7	0.734
5.8-2	5.3.1-3	0.733
5.8-2	5.3.1-2.2	0.728
5.8-2	5.1.2-5	0.728
5.8-2	5.2.2-1	0.727
5.8-2	5.2.1-4.1	0.726
5.8-2	5.5.1-1	0.726
5.8-2	5.1.3-1.1	0.721
5.8-2	5.1.1-2	0.720
5.8-2	5.2.2-3	0.718
5.8-2	5.4.2-3	0.717
5.8-2	5.1.2-2	0.715
5.8-2	5.1.3-2	0.715
5.8-2	5.1.4-4	0.714
5.8-2	5.2.2-2	0.713
5.8-3	5.3.1-1	0.793
5.8-3	5.2.2-4	0.769
5.8-3	5.3.1-2.2	0.761

EN 303 645	EN 304 223	Score
5.8-3	5.2.1-1	0.760
5.8-3	5.2.4-1.1	0.760
5.8-3	5.2.3-2.2	0.759
5.8-3	5.3.1-2	0.758
5.8-3	5.1.2-3	0.755
5.8-3	5.2.4-1	0.749
5.8-3	5.2.1-4	0.745
5.8-3	5.1.4-2	0.740
5.8-3	5.4.2-3	0.740
5.8-3	5.2.4-2	0.740
5.8-3	5.1.4-5	0.737
5.8-3	5.2.2-6	0.735
5.8-3	5.1.2-4	0.735
5.8-3	5.1.2-7	0.734
5.8-3	5.4.1-1	0.734
5.8-3	5.1.4-1	0.734
5.8-3	5.1.3-2	0.728
5.8-3	5.2.3-1	0.727
5.8-3	5.1.2-1	0.726
5.8-3	5.2.5-1	0.725
5.8-3	5.1.3-1	0.724
5.8-3	5.1.3-3	0.724
5.8-3	5.2.3-2.1	0.723
5.8-3	5.2.3-4	0.722
5.8-3	5.2.4-2.1	0.722
5.8-3	5.2.3-2	0.721
5.8-3	5.2.1-4.2	0.719
5.8-3	5.1.4-3	0.718
5.8-3	5.2.1-4.1	0.717
5.8-3	5.1.2-2	0.716
5.8-3	5.2.4-3	0.715
5.8-3	5.2.5-3	0.714
5.9-1	5.1.2-2	0.750
5.9-1	5.2.1-3	0.720
5.9-1	5.3.1-3	0.717
5.9-2	5.1.2-2	0.719